

Master of Science (M.Sc.)

”Business Informatics“



– Module catalog –

Academic Year
HWS 2026 / FSS 2027

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Foreword

This document describes the courses that will be offered in academic year HWS 2026/ FSS 2027 for students studying M. Sc. Business Informatics (Examination Regulations for the Master's Program from 12.12.2017). You can find the Examination Regulations on the website of the Student Services (Studienbüros):

<https://www.uni-mannheim.de/en/examinations/rechtliche-grundlagen/examination-regulations/>

It is possible that additional courses will be made available during the course of the academic year. These will be published in an appendix available on the following web page:

<https://www.wim.uni-mannheim.de/studium/studienorganisation/m-sc-business-informatics/>

A. Overview

		ECTS
Fundamentals Computer Science	Three “Computer Science Fundamentals” courses	18
Fundamentals Business Administration	Courses from the module catalog of the “Mannheim Master in Management”	18 (at least)
Specialization Courses	Specialization Courses	36 (at least)
Projects and Seminars	Team Project, Scientific Research and Seminars	18
Master’s Thesis	Six-month-long written academic assignment	30
Total		120

Abbreviations:

- HWS (Herbst-/Wintersemester): Course is offered in the respective Fall semester.
- FSS (Frühjahrs-/Sommersemester): Course is offered in the respective Spring semester.
- HWS/FSS: course is offered both in Fall semester and Spring semester.

Please note: the module descriptions of all IS courses can be found in the module catalog of the “Mannheim Master in Management” which can be found here:

<https://www.bwl.uni-mannheim.de/modulkatalog-mmm/>

General constraints:

See Examination Regulations on the website of the Student Services (Studienbüros):

<https://www.uni-mannheim.de/pruefungen/>

B. Fundamentals

1. Overview

Fundamentals Computer Science

Module no.	Name of Module	Offered	Language	ECTS	Page
CS 500	Advanced Software Engineering	HWS	E	6	5
CS 530	Database Systems II	FSS	E	6	7
CS 550	Algorithmics	FSS	E	6	9
CS 560	Large-Scale Data Management	HWS	E	6	11
CS 652	Data Security and Privacy	FSS	E	6	13
IE 500	Data Mining	HWS/FSS	E	6	15
IE 560	Foundations of Artificial Intelligence - Reasoning and Decision Making	HWS	E	6	17

Fundamentals Business Administration

All 5XX (5XXX) and 6XX (6XXX) courses from the following areas:

- Accounting and Taxation (ACC, TAX)
- Banking, Finance and Insurance (FIN)
- Management (MAN)
- Marketing (MKT)
- Operations Management (OPM)

These courses are listed in the module catalog of the “Mannheim Master in Management”:

<https://www.bwl.uni-mannheim.de/modulkatalog-mmm/>

The following modules are excluded from enrollment:

MAN 605 (MAN 6050), MAN 608 (MAN 6080), MAN 646 (MAN 6460), MAN 647 (MAN 6470), MAN 648 (MAN 6480), MAN 654 (MAN 6540), MAN 655 (MAN 6550), MAN 656 (MAN 6560), MAN 699 (MAN 6990), MKT 575 (MKT 5750), MKT 580 (MKT 5800), MKT 622 (MKT 6220), MKT 623 (MKT 6230), MKT 625 (MKT 6250)

2. Detailed descriptions

Fundamentals Computer Science

CS 500: Advanced Software Engineering

Module number	CS 500
Title	Advanced Software Engineering
Form of module	Lecture, exercise and tutorial
Type of module	Computer Science Fundamental
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (2+2 SWS) Self-study: 112 h per semester, including • 28 h: Preparation and studying of lectures/tutorials • 84 h: Preparation of the exams
Prerequisites	Knowledge of Java and/or Python
Aim of module	The course introduces the fundamental concepts and practices of software testing and software experimentation / evaluation • Principles of Testing • Test Coverage Criteria • Test specification techniques • Model-based testing • Principles of Experimentation • Use of Software Observatoriums
Learning outcomes and qualification goals	Professional competence: • Students will be familiar with principles, approaches and tools for testing software and the methods and empirically evaluating software products, methods and tools (MK1, MK2) Methodological competence: • Students will have the ability to judge, select and apply traditional or non-traditional test tools and techniques to evaluate software quality, as well as to design and perform software experiments (MF1, MF3) Personal competence: • Students will have improved skills in analytical thinking and applying theoretical knowledge to solve practical problems, especially in software testing and experimentation (MKO3)
Media	Lecture notes, presentations, tool demonstrations, software, data sets
Literature	Paul Ammann & Jeff Offutt., <i>Introduction to Software Testing</i> , Cambridge University Press, January 2008
Methods	Lectures, weekly exercises, experimentation with different systems
Form of assessment	Written examination

Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Colin Atkinson
Duration of module	1 semester
Further modules	Model-Driven Development, Generative Software Engineering
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	1st/ 2nd semester

CS 530: Database Systems II

Modulnummer	CS 530
Titel	Datenbanksysteme II / <i>Database Systems II (DBSII)</i>
Form der Veranstaltung	Vorlesung mit Übung
Typ der Veranstaltung	Vertiefung
Modulniveau	Master
ECTS	6
Arbeitsaufwand	Präsenzstudium: 56 h pro Semester (4 SWS) Eigenstudium: 56 h pro Semester, davon • 28 h: Vor- und Nachbereitung der Veranstaltung und freies Selbststudium • 28 h: Vorbereitung für die Prüfung
Vorausgesetzte Kenntnisse	PI2, Algorithmen und Datenstrukturen, DBSI, C, C++
Lehrinhalte	Implementierung von Hauptspeicherdatenbanksystemen
Lern- und Kompetenzziele	Fachkompetenz: • Effiziente C/C++ Programme entwickeln • Physische Algebra • Implementierung von Indexstrukturen (MK1, MK3) Methodenkompetenz: • Analyse komplexer Algorithmen (MF1) Personale Kompetenz: • Präzises Analysieren (MKO1, MKO3)
Medienformen	Präsentationen mit Tafelanschrieb
Begleitende Literatur	Skript
Lehr- und Lernmethoden	Vorlesung
Art der Prüfungsleistung	Schriftliche Prüfung
Prüfungsvorleistung	-
Prüfungsdauer	90 Minuten
Sprache	Englisch
Angebotsturnus	FSS
Modulverantwortliche/r	Prof. Dr. Guido Moerkotte
Dauer des Moduls	1 Semester
Weiterführende Module	-

Verwendbarkeit	M.Sc. Wirtschaftsinformatik, Lehramt Informatik
Einordnung in Fachsemester	1./ 2./ 3. Fachsemester

CS 550: Algorithmics

Module number	CS 550
Title	Algorithmics
Form of module	Lecture with exercise
Type of module	Fundamental in Computer Science
Level	Master
ECTS	6
Workload	Attendance: 56 h per semester (4 h per week) Self-study: 112 h per semester, including • 28 h: Preparation and reworking of lectures/tutorials • 84 h: Preparation of the exams
Prerequisites	Practical Informatics I, Algorithms and Data Structures, Linear Algebra, Statistics
Aim of module	The lecture deals with the design and the analysis of algorithms for various practically relevant computational problems and with methods for analyzing the complexity of certain problems. In particular, we will learn methods of formalizing discrete optimization problems and designing algorithms for them on the basis of analyzing the structure of these problems. Moreover, we will learn techniques for proving the correctness and estimating the running time of these algorithms. In the second part of the lecture, we will deal with the theory of NP-completeness with gives evidence that certain highly relevant problems do not have efficient algorithms. During the lecture we will derive algorithms and complexity-theoretic results for the following computational problems: • Shortest path problems and shortest round tour problems • Linear optimization problems • Flow problem • Matching problems • Satisfiability problems • Discrete linear optimization problems
Learning outcomes and qualification goals	Professional competence: • The students know efficient algorithms and the most important complexity-theoretic results for a number of computational problems which are highly relevant in practice (MK1, MK2) Methodological competence: • The students learn to formalize informally specified computational problems • They learn to analyze their structure with the goal to design efficient algorithms • They learn to prove the correctness • They learn to analyze the running time of given algorithms • They learn to prove the NP-completeness of certain problems (MF1, MF3)

	Personal competence: • Training of analytical, focussed and precise thinking • Development of abstraction abilities and the ability to transfer theoretical knowledge for solving practical problems, especially in the field of operations research • Increasing the sensitivity for the complexity and the efficient solvability of computational problems, especially through dealing with the theory of NP-completeness (MF1, MKO3)
Media	Lectures Slides and Exercises
Literature	• Cormen, Leiserson, Rivest, Stein: <i>Introduction to Algorithms</i>, 3rd edition • Shimon Even: <i>Graph Algorithms</i> • Lovasz, Plummer: <i>Matching Theory</i> • <i>Handbooks on Operations Research and Management Science</i>, Volume 7 (Editors: Ball, Magnati, Monma, Nemhauser) • J. Toran: <i>Das Erfüllbarkeitsproblem SAT</i>, Lehmann Media, 2012
Methods	Reworking of lectures and tutorials, self-studies with literature, solving exercises at home and in cooperation with other students at the tutorials
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr. Matthias Krause
Duration of module	1 semester
Further modules	CS 651 - Cryptography II
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Mathematics in Business and Economics, M.Sc. Economic and Business Education, Teacher Education Informatics, M.Sc. Mathematics
Semester	1st/ 2nd semester

CS 560: Large-Scale Data Management

Module number	CS 560
Title	Large-Scale Data Management
Form of module	Lecture with exercise
Type of module	Computer Science Fundamental
Level	Master
ECTS	6
Workload	Hours in presence: 42 h per semester (2+2 SWS) Exercises and self-study: 119 h per semester
Prerequisites	Database systems, algorithms & data structures, Java/Python programming
Aim of module	This course introduces the fundamental concepts and computational paradigms of large-scale data management and Big Data. This includes methods for storing, updating, querying and analyzing large dataset as well as for data-intensive computing. The course covers concept, algorithms and system issues; accompanying exercises provide hands-on experience. Topics include: • Parallel and distributed databases • Parallel database design • Distributed query processing • Big data processing (including MapReduce, Spark, Streaming) • Distributed transactions • NoSQL databases
Learning outcomes and qualification goals	Professional competence: • Students will acquire knowledge about methods and systems for working with large datasets and data-intensive computing (MK1, MK2) Methodological competence: • Be able to judge, select and use traditional and modern data management systems for a given data intensive task • Be able to solve computational problems involving large datasets (MF1) Personal competence: • Study independently (MKO3)
Media	Slide set, black board, exercise sheets, datasets, software
Literature	• T. Özsu, P. Valduriez: <i>Principles of Distributed Database Systems</i>, Springer, 4th ed., 2020 • H. Garcia-Molina, J. D. Ullman, J. Widom: <i>Database Systems: The Complete Book</i>, Prentice Hall, 2nd ed., 2008 • More in lecture notes
Methods	Lecture, weekly exercise, experimentation with different systems
Form of assessment	Written examination

Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Rainer Gemulla
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	1st/ 2nd semester

CS 652: Data Security and Privacy

Module number	CS 652
Title	Data Security and Privacy
Form of module	Inverted classroom with exercises
Type of module	Fundamental in Computer Science
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 112 h per semester
Prerequisites	There are no formal prerequisites but knowledge in cryptography or IT-security is recommended, e.g., by attending the lectures “Kryptographie I” or “Selected Topics in IT-Security”
Aim of module	Nowadays, more and more user-related data is collected and processed – either willingly as in the context of Facebook or Payback or possibly unconsciously as in the case of the Internet of Things. The aim of the module is to raise awareness, in particular with respect to privacy risks. This is done by showing various threats. Possible examples are traces left on the Internet such as the use of cookies or browser fingerprinting but also leakage in the context of machine learning. In addition, a variety of privacy enhancing technologies will be discussed, possibly such as anonymization of data, differential privacy, or the use of dedicated cryptographic schemes
Learning outcomes and qualification goals	Professional competence: - Students will acquire the knowledge to identify security and privacy threats and to select and use appropriate countermeasures (MK2) Methodological competence: - Successful participants will be able to understand, to select, apply and evaluate the most appropriate techniques for a variety of different privacy-sensitive scenarios - In particular they are able to realize possible risks in new scenarios and to transfer given solutions to these (MK1) Personal competence: - The analytic, concentrated and precise thinking of the students is trained - By the independent treatment of applications, e.g. in the course of the exercises, their abstraction capacity is further developed and the transfer of the learned material to related questions is trained (MF1, MKO3)
Media	Video recordings, annotated lecture slides
Literature	Will be announced in the lecture

Methods	Reworking the lecture and studying the relevant literature in self-study. During the lecture: discussing questions and ideas and working together on concrete examples. Solving exercises in self-study and in practice in cooperation with fellow students
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr. Frederik Armknecht
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science, M.Sc. Business Informatics, Teacher Education Informatics, M.Sc. Mathematics
Semester	1st/ 2nd/ 3rd semester

IE 500: Data Mining

Module number	IE 500
Title	Data Mining
Form of module	Lecture with exercise and project
Type of module	Business Informatics Fundamental
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 98 h per semester, including • 70 h: Pre- and post-lecture studying and revision • 28 h: Examination preparation
Prerequisites	Foundations of Statistics, Practical Informatics I
Aim of module	The course provides an introduction to advanced data analysis techniques as a basis for analyzing business data and providing input for decision support systems. The course will cover the following topics: • Goals and Principles of Data Mining • Data Representation and Preprocessing • Classification • Regression • Cluster Analysis • Association Analysis • Practical Data Mining using Python
Learning outcomes and qualification goals	Professional competence: • Students will acquire basic knowledge of the techniques, opportunities and applications of data mining (MK1, MF1) Methodological competence: • Successful participants will be able to identify opportunities for applying data mining in an enterprise environment, select and apply appropriate techniques, and interpret the results • Project organization skills (MK2, MF3, MF4, MKO1) Personal competence: • Team work skills • Presentation skills (MKO2, MF2)
Media	Slide set, exercise sheets, Jupyter notebooks, data sets for the exercises, lecture videos
Literature	• Pang-Ning Tan, Michael Steinback, Vipin Kumar: <i>Introduction to Data Mining</i>, 2nd Edition, Pearson • Aurélien Géron: <i>Hands-on Machine Learning with Scikit-Learn, Keras & TensorFlow</i>, 3rd Edition, O'Reilly

Methods	The course consists of a lecture together with accompanying practical exercises as well as student team projects. In the exercises the participants will gather initial expertise in applying state of the art data mining tools on realistic data sets. The group projects take place in the last third of the term. Within the projects, students implement advanced data mining projects of personal choice and report about the results of their projects in the form of a written report as well as an oral presentation
Form of assessment	• Written examination (75%) • Project report (20%) • Oral project presentation (5%)
Admission requirements for assessment	-
Duration of assessment	• 60 minutes (written examination) • 5-11 pages (project report) • 5-15 minutes (oral project presentation)
Language	English
Offering	HWS/FSS
Person in charge	Prof. Dr. Heiko Paulheim, Prof. Dr. Christian Bizer, Dr. Sven Hertling
Duration of module	1 semester
Further modules	• IE 675b - Machine Learning • IE 678 - Deep Learning • IS 661 (IS 6610) - Text Analytics • CS 646 - Higher Level Computer Vision
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	1st/ 2nd semester

IE 560: Foundations of Artificial Intelligence - Reasoning and Decision Making

Module number	IE 560
Title	Foundations of Artificial Intelligence - Reasoning and Decision Making
Form of module	Inverted classroom
Type of module	Business Informatics Fundamental
Level	Master
ECTS	6
Workload	Hours in presence: 50 h per semester (2+2 SWS) Self-study: 120 h per semester, including • 80 h: Pre- and post-lecture studying and revision • 40 h: Examination preparation
Prerequisites	Linear Algebra, Basic Probability Theory, Basic Knowledge of Propositional Logic
Aim of module	The course provides an introduction to methods for AI-based decision making as a basis for the design of automated decision making and decision support systems. The course will cover the following topics: • Decision Problems and Bounded Rationality • Propositional Logic • Probability Theory • Probabilistic Graphical Models • Utility and Decision Theory • Game Theory • Markov Decision Processes • Reinforcement Learning
Learning outcomes and qualification goals	Professional competence: • Students will acquire basic knowledge of the techniques, opportunities and applications of AI-based decision making (MK1, MF1) Methodological competence: • Successful participants will be able to identify opportunities for decision support in an enterprise environment, select and apply appropriate techniques, and interpret the results (MK2, MF3, MF4, MKO1) Personal competence: -
Media	Lecture videos, slide set, exercise sheets, software tools
Literature	S. Russel and P. Norvig: <i>AI: A Modern Approach</i> (3rd Edition), 2010 (selected sections)
Methods	The course consists of a lecture accompanied by practical homework and case studies. In the lecture, the students' basic concepts and methods of decision theory will be explained both in theory and using concrete examples. The students will practice and test their knowledge acquired in the lecture in homework assignments. Within the case studies, students model real world decision problems and try to solve them optimally using methods from the lecture

Form of assessment	Written examination
Admission requirements for assessment	Midterm exam: 45 minutes (propositional logic and probability theory)
Duration of assessment	90 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Heiner Stuckenschmidt
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Social Data Science
Semester	1st/ 2nd semester

C. Specialization Courses

1. Overview

CS-Courses:

Module no.	Name of Module	Offered	Language	ECTS	Page
CS 600	Model Driven Development	HWS	E	6	22
CS 606	Foundations of Artificial Intelligence: Search and Problem Solving	HWS	D	6	24
CS 630	Generative Software Engineering	FSS	E	6	26
CS 646	Higher Level Computer Vision	HWS	E	6	28
CS 647	Image Processing	FSS	E	6	30
CS 651	Cryptography II	HWS	E	6	32
CS 664	Blockchain Security	HWS	E	6	34
CS 668	Generative Computer Vision Models	FSS	E	6	36

IE-Courses:

Module no.	Name of Module	Offered	Language	ECTS	Page
IE 630	Anfrageoptimierung	FSS	D	6	38
IE 650	Knowledge Graphs	HWS	E	6	40
IE 655	Industrial Applications of Artificial Intelligence	FSS	E	9	42
IE 670	Web Data Integration	HWS	E	3	44
IE 675b	Machine Learning	HWS	E	9	46
IE 678	Deep Learning	FSS	E	6	48
IE 679	AI Deployment and Operations	HWS	E	3	50
IE 683	Web Data Integration Project	HWS	E	3	52
IE 685	Large Language Models and Agents	FSS	E	3	54
IE 686	Large Language Models and Agents Project	FSS	E	3	56
IE 692	Advanced Process Mining	FSS	E	6	58
IE 695	Reinforcement Learning	HWS	E	6	60
IE 696	Advanced Methods in Text Analytics	FSS	E	6	62
IE 698	Foundations and Applications of Digital Health Technologies	FSS	E	3	64

IE 699	Co-creating digital health applications with design methodology	FSS	E	6	66
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IS-Courses:

Module no.	Name of Module	Offered	Language	ECTS	Page
IS 512 (IS 5120)	IT Management in the Digital Age	MMM*	E	6	MMM*
IS 513 (IS 5130)	Applied IT Management in the Digital Age	MMM*	E	6	MMM*
IS 540 (IS 5400)	Management of Enterprise Systems	MMM*	E	6	MMM*
IS 541 (IS 5410)	Methods and Theories in Information Systems (ManTIS)	MMM*	E	6	MMM*
IS 603 (IS 6030)	IT Leadership in Organizations	MMM*	E	6	MMM*
IS 607 (IS 6070)	Digital Innovation	MMM*	E	6	MMM*
IS 609 (IS 6090)	AI and Technology Strategy	MMM*	E	6	MMM*
IS 611 (IS 6110)	Advanced Topics in Large Language Models	MMM*	E	6	MMM*
IS 612 (IS 6120)	Product Experimentation and Analytics	MMM*	E	6	MMM*
IS 613 (IS 6130)	Applied Project in Design Thinking and Lean Software Development	MMM*	E	6	MMM*
IS 614 (IS 6140)	Corporate Knowledge Management	MMM*	E	6	MMM*
IS 615 (IS 6150)	Enterprise Cloud Design and Development	MMM*	E	6	MMM*
IS 617 (IS 6170)	Large Language Models for the Economic and Social Sciences	MMM*	E	6	MMM*
IS 629 (IS 6290)	Agile Software Product Management and Design	MMM*	E	6	MMM*
IS 661 (IS 6610)	Text Analytics	MMM*	E	6	MMM*

* For a detailed description, please see the module catalogue of the respective following degree programs:

MMM: M.Sc. Mannheim Master in Management, <https://www.bwl.uni-mannheim.de/en/module-catalog-mmm/>

** Prerequisite: No completed exam in IE 676

International Courses and other Specialization Courses:

Module no.	Name of Module	Offered	Language	ECTS	Page
BI 656	International Course	HWS/FSS	-	max. 18	69
DS 203	Responsible AI: Conceptual Foundations, Methods and Applications	HWS	E	6	70
MAB 519	Reinforcement Learning	FSS	E	10	72
MAC 404	Lineare Optimierung	HWS	D	8	74

2. Detailed descriptions

CS-Courses:

CS 600: Model Driven Development

Module number	CS 600
Title	Model-Driven Development
Form of module	Lecture with accompanying tutorials
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (2+2 SWS) Self-study: 124 h per semester, including • 32 h: Pre- and post-lecture studying and revision • 60 h: Tutorial exercises • 32 h: Examination preparation
Prerequisites	CS 500: Advanced Software Engineering
Aim of module	The course focuses on the principles, practices and tools involved in advanced model-driven development. This includes established modelling standard languages (e. g. UML, ATL, OCL) and modelling infrastructures (e. g. MOF, EMF, ...) as well as leading edge, state-of-the-art modelling technologies (e. g. LML, PLM . . .). Key topics addressed include: • Principles of Model-Driven Development • Meta-Modeling • Domain-Specific Languages • Model Constraint and Transformation languages • Multi-View/Multi-Paradigm Modeling • Multi-Level Modeling • Ontology Modeling • Enterprise Architecture Modeling
Learning outcomes and qualification goals	Professional competence: • Students will be familiar with the accepted best practices and technologies used in mainstream model-driven development as well as state-of-the-art modeling technologies emerging from research institutions (MK1, MK2) Methodological competence: • Students will know how to apply modeling technologies in real-world projects (MF1, MF3) Personal competence: • Students will have the capability to analyse, understand and model complex systems (MKO1)
Media	Printed Lecture notes, Presentations, Tool Demonstrations

Literature	Jos B. Warmer and Anneke G. Kleppe, <i>The Object Constraint Language: Getting Your Models Ready for MDA</i> , Addison-Wesley Object Technology Series, 2003
Methods	The course consists of • Lectures that introduce, explain and exemplify the topics covered in the course • Individual projects in which each student develops a small system utilizing the topics covered in the course • Tutorials involving practical exercises and discussions of the individual projects
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Colin Atkinson
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	2nd/ 3rd/ 4th semester

CS 606: Foundations of Artificial Intelligence - Search and Problem Solving

Modulnummer	CS 606
Titel	Foundations of Artificial Intelligence: Search and Problem Solving
Form der Veranstaltung	Vorlesung mit Übung
Typ der Veranstaltung	Vertiefung Informatik
Modulniveau	Master
ECTS	6
Arbeitsaufwand	Präsenzstudium: 56 h pro Semester (4 SWS) Eigenstudium: ca. 98 h pro Semester, davon • 70 h: Vor- und Nachbereitung der Veranstaltung und freies Selbststudium • 28 h: Vorbereitung für die Prüfung, z.B. Prüfungs-/ Seminarabschlussarbeits- und Präsentationsvorbereitung
Vorausgesetzte Kenntnisse	Formale Grundlagen der Informatik (insbes. Komplexität, Graphen und Logik), Algorithmen & Datenstrukturen, Programmierpraktikum I & II
Lehrinhalte	• Problemeigenschaften und Problemtypen • Problemlösen als Suche, Anwendung im Bereich Spiele • Constraintprobleme und deren Lösung • Logische Constraints • Entscheidungsbaumlernen
Lern- und Kompetenzziele	Fachkompetenz: • Ziele und Grundlagen der Künstlichen Intelligenz • Suchverfahren als universelle Problemlösungsverfahren • Problemkomplexität und Heuristische Lösungen • Eigenschaften und Zusammenhang zwischen unterschiedlichen Suchverfahren (BK7) Methodenkompetenz: • Beschreibung konkreter Aufgaben als Such-, Constraint- oder Planungsproblem • Implementierung unterschiedlicher Suchverfahren und Heuristiken (BF1, BKO1) Personale Kompetenzen: -
Medienformen	Lehrbuch, Vorlesung, Präsentationen, Tafel
Begleitende Literatur	Russel and Norvig: <i>Artificial Intelligence - a modern approach</i> . Prentice Hall. 3rd Edition 2010
Lehr- und Lernmethoden	Vorlesung, praktische Übungen, eigenständige Bearbeitung von Übungsaufgaben
Art der Prüfungsleistung	Schriftliche Prüfung
Prüfungsvorleistung	• 1-2 Mini-Projekte erfolgreich bearbeitet und dokumentiert • Mindestens 60% aller Aufgabenpunkte aufsummiert

Prüfungsdauer	90 Minuten
Sprache	Deutsch/Englisch
Angebotsturnus	HWS
Modulverantwortliche/r	Prof. Dr. Heiner Stuckenschmidt
Dauer des Moduls	1 Semester
Weiterführende Module	SM 442 Bachelorseminar Prof. Stuckenschmidt
Verwendbarkeit	M.Sc. Wirtschaftsinformatik, B.Sc. Wirtschaftsinformatik, M.Sc. Wirtschaftspädagogik, Lehramt Informatik, Beifach Angewandte Informatik, M.Sc. Data Science
Einordnung in Fachsemester	5./ 6. Fachsemester im Bachelor, 1./ 2./ 3. Fachsemester im Master

CS 630: Generative Software Engineering

Module number	CS 630
Title	Generative Software Engineering
Form of module	Lecture, exercises, tutorials and projects
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (2+2 SWS) Self-study: 112 h per semester, including • 28 h: Preparation and studying of lectures/tutorials • 84 h: Preparation of the exams
Prerequisites	• Software Testing and Experimentation • Knowledge of Java and/or Python
Aim of module	The course introduces the fundamental principles, practices and applications of generative software engineering (GSE) as well as the supporting models, tools and services: • Software Recommendation and Reuse • Code Search Engines • Principles of GSE • Existing GSE models, tools and services • Defining GSE “problems” • Applications of GSE • Evaluation and comparison of GSE models, tools and services
Learning outcomes and qualification goals	Professional competence: • Students will be familiar with the principles, practices and applications of GSE as well as supporting models, tools and services (MK1, MK2) Methodological competence: • Students will have the ability to judge, select and apply GSE techniques and tools in practical software engineering • They will be able to understand academic GSE publications and perform research in GSE (MF1, MF2, MF3) Personal competence: • Students will have improved skills in analytical thinking and applying theoretical knowledge to solve practical problems, especially in the use of AI to enhance software engineering (MKO1, MKO3)
Media	Students acquire knowledge of the principles, practices and applications of GSE, including supporting models, tools and services and methods, through lectures and tutorials. The students apply their knowledge individually and in team projects and report their findings in written reports and oral presentations
Literature	-
Methods	Lectures, weekly exercises, experimentation with different systems

Form of assessment	• Project reports (70%) • Oral presentations (30%)
Admission requirements for assessment	-
Duration of assessment	-
Language	English
Offering	FSS
Person in charge	Dr. Marcus Kessel
Duration of module	1 semester
Further modules	Model-Driven Development, Software Testing and Experimentation
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	2nd/ 3rd/ 4th semester

CS 646: Higher Level Computer Vision

Module number	CS 646
Title	Higher Level Computer Vision
Form of module	Lecture with exercise
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 98 h per semester, including • 70 h: Lecture/exercises • 28 h: Examination preparation
Prerequisites	Basic skills in linear algebra, basic knowledge in Python and PyTorch
Aim of module	• Point Features and point matching • Object Identification • Deep Learning for Computer Vision • Object Detection • Image Segmentation • Optical Flow • Video and Motion Segmentation
Learning outcomes and qualification goals	Professional competence: • The students have a detailed understanding of Computer Vision techniques • They can evaluate given Computer Vision algorithms (MK1, MK2, MF1, MF3) Methodological competence: • Students understand the technical basis of Computer Vision algorithms • They can explain the discussed methods and implement them (MF1, MF2, MF3) Personal competence: • Understanding complex Computer Vision problems • Thorough judgment in the design and use of methods • Can work efficiently in a team (MKO1, MKO2)
Media	Exercise sheets and lecture slides are available online
Literature	• Goodfellow et al.: <i>Deep Learning</i>, MIT Press, 2016 https://www.deeplearningbook.org/ • R. Szeliski: <i>Computer Vision: Algorithms and Applications</i>, Springer, 2010. ISBN: 978-1-84882-934-3. http://szeliski.org/Book/ • D. Forsyth, J. Ponce: <i>Computer Vision: A Modern Approach</i>, Prentice Hall, 2nd ed., 2012. ISBN: 978-0136085928. • R. Hartley, A. Zisserman: <i>Multiple View Geometry in Computer Vision</i>, Cambridge University Press, 2nd ed., 2004

Methods	Lecture, weekly exercise, book studies, implementation of algorithms, visualization of results
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr.-Ing. Margret Keuper
Duration of module	1 semester
Further modules	Image Processing, Generative Computer Vision Models
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	1st/ 2nd/ 3rd semester

CS 647: Image Processing

Module number	CS 647
Title	Image Processing
Form of module	Lecture with exercise (partially online)
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 98 h per semester, including • 70 h: Lecture/exercises • 28 h: Examination preparation
Prerequisites	Basis skills in linear algebra, basis knowledge in python
Aim of module	• Introduction to Imaging (human visual system, optics, sensors) • Noise and basic operations (convolution, correlations, gradients) • Energy minimization • Variational Methods • Fourier theory and wavelet theory • Image Denoising • Segmentation • Stereo Vision
Learning outcomes and qualification goals	Professional competence: • The students have a detailed understanding of image and video processing techniques • They can evaluate given image processing algorithms (MK1, MK2, MF1, MF3) Methodological competence: • Students understand the technical basis of image processing algorithms • They can explain the discussed methods and implement them (MF1, MF2, MF3) Personal competence: • Understanding complex Image Processing problems • Thorough judgment in the design and use of methods • Can work efficiently in a team (MKO1, MKO2)
Media	Exercise sheets and lecture slides available online
Literature	• R. Szeliski: Computer Vision Algorithms and Applications, Springer, 2010. ISBN: 978-1-84882-934-3. (Online available: http://szeliski.org/Book/) • D. Forsyth, J. Ponce: Computer Vision: A Modern Approach, Prentice Hall, 2nd edition, 2012. ISBN: 978-0136085928
Methods	Lecture, weekly exercise, book studies, implementation of algorithms, visualization of results
Form of assessment	Written examination

Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr.-Ing. Margret Keuper
Duration of module	1 semester
Further modules	Higher Level Computer Vision, Generative Computer Vision Models
Programs	B.Sc. Business Informatics, M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	1st/ 2nd/ 3rd semester

CS 651: Cryptography II

Module number	CS 651
Title	Cryptography II
Form of module	Inverted classroom
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 112 h per semester
Prerequisites	Even though the lecture deepens and continues topics discussed in “CS 404 Cryptography I”, it is not a prerequisite to have attended this lecture. The lecture “Cryptography II” can be studied without any prior knowledge on cryptography - all necessary basics are shortly recapitulated
Aim of module	The goal of this lecture is to present and discuss important scientific concepts from modern cryptography. Possible topics are: • Security Definitions: How can the security of cryptographic schemes formally defined? • Proofs of Security: How can the security of cryptographic schemes be proven (based on precise assumptions)? • Cryptanalysis: What are the established techniques to analyze cryptographic mechanisms? • Elliptic Curves • Zero-Knowledge Proofs
Learning outcomes and qualification goals	Professional competence: • With the help of current techniques and theories of modern cryptography, the students can assess the security of cryptographic processes and assess security statements accordingly • They can identify security goals and use appropriate techniques that could not be dealt with in “CS 404 Cryptography I” (MK2) Methodological competence: • The students can select and use suitable methods for the security analysis of cryptographic processes. This includes, for example, the choice of the appropriate security model, proof of security based on clearly specified assumptions and the analysis of given procedures • The students can understand and assess the security arguments for existing procedures and to transfer them to new ones. • They can use techniques and protocols to achieve security goals that were not yet possible with the methods discussed in “CS 404 Cryptography I” (MK1)

	Personal competence: • The analytic, concentrated and precise thinking of the students is trained • By the independent treatment of applications, e.g. in the course of the exercises, their abstraction capacity is further developed and the transfer of the learned material to related questions is trained (MF1, MKO3)
Media	Video recordings, annotated lecture slides
Literature	Jonathan Katz, Yehuda Lindell: <i>Introduction to Modern Cryptography: Principles and Protocols</i> , Chapman and Hall/CRC, 2007
Methods	Reworking the lecture and studying the relevant literature in self-study. During the lecture: discussing questions and ideas and working together on concrete examples. Solving exercises in self-study and in practice in cooperation with fellow students
Form of assessment	Oral examination
Admission requirements for assessment	-
Duration of assessment	30 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Frederik Armknecht
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science, M.Sc. Business Informatics, Teacher Education Informatics, M.Sc. Mathematics
Semester	1st/ 2nd/ 3rd semester

CS 664: Blockchain Security

Module number	CS 664
Title	Blockchain Security
Form of module	Inverted classroom with exercises
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 112 h per semester
Prerequisites	There are no formal prerequisites but knowledge in cryptography and/or IT-security is recommended, e.g., by attending the lectures “Kryptographie I” or “Selected Topics in IT-Security”
Aim of module	Besides cryptocurrencies, blockchain technology has become increasingly important for a variety of use cases such as finance, supply chain management, digital identity, and decentralized applications. The main difference to previous technologies is that it promises secure and reliable data storage and consensus in a trustless environment. Still, we observe a lack of understanding on the security properties and limitations of blockchain technology. The aim of this lecture is to close this gap by focusing on the security of blockchain technology. The course will equip students with a solid understanding of blockchains, their design principles, underlying technologies, and cryptographic primitives. Bitcoin, Monero and Ethereum will be discussed in greater detail and a substantial part of the course will be devoted to security issues and possible attacks
Learning outcomes and qualification goals	Professional competence: - Students will acquire profound knowledge of Blockchain technology as well as the skills to critically examine the security of Blockchain-based systems (MK1, MK2) Methodological competence: - Successful participants will be able to understand and evaluate the different ways in which different Blockchain systems try to achieve security - They will also be able to identify where, why and how these security measures are broken for both, current and new systems (MKO3) Personal competence: - The analytic, concentrated and precise thinking of the students is trained - As multiple different but related Blockchains are discussed, their abstraction capacity is further developed and the transfer of the learned concepts to related questions is trained (MF1)
Media	Video recordings, annotated lecture slides
Literature	Will be announced in the lecture

Methods	Reworking the lecture and studying the relevant literature in self-study. During the lecture: discussing questions and ideas and working together on concrete examples. Solving exercises in self-study and in practice in cooperation with fellow students
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Frederik Armknecht
Duration of Module	1 Semester
Further Modules	-
Programs	M.Sc. Data Science, M.Sc. Business Informatics, Teacher Education Informatics, M.Sc. Mathematics, M.Sc. Mathematics in Business and Economics
Semester	1st/ 2nd/ 3rd semester

CS 668: Generative Computer Vision Models

Module number	CS 668
Title	Generative Computer Vision Models
Form of module	Lecture with exercise
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 98 h per semester, including • 70 h: Lecture/exercises • 28 h: Examination preparation
Prerequisites	Basis skills in linear algebra, basis knowledge in python/pytorch
Aim of module	• Introduction to Clustering and Unsupervised Learning • Introduction to Generative Models • Autoregressive Models for Image Generation • (Normalizing) Flow • Latent Variable Models • Latent Space Visualizations • Generative Adversarial Models • Diffusion Models • Multi-Modal Conditioning
Learning outcomes and qualification goals	Professional competence: • The students have a detailed understanding of image generation techniques, latent variable models and their evaluations (MK1, MK2, MF1, MF3) Methodological competence: • Students understand the technical basis of generative models • They can explain the discussed methods and implement them (MF1, MF2, MF3) Personal competence: • Understanding image generative models, their working principles and training procedures • Thorough judgment in the design and use of methods • Students can work efficiently in a team (MKO1, MKO2)
Media	Exercise sheets and lecture slides available online
Literature	Ian Goodfellow and Yoshua Bengio and Aaron Courville, <i>Deep Learning</i> , An MIT press book, 2016
Methods	Lecture, weekly exercise, book studies, implementation of algorithms, visualization of results
Form of assessment	Written examination

Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr.-Ing. Margret Keuper
Duration of module	1 semester
Further modules	Higher Level Computer Vision, Genarative Computer Vision Models
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	1st/ 2nd/ 3rd semester

IE-Courses:

IE 630: Anfrageoptimierung

Modulnummer	IE 630
Titel	Anfrageoptimierung / <i>Query Optimization</i>
Form der Veranstaltung	Vorlesung mit Übung
Typ der Veranstaltung	Vertiefung
Modulniveau	Master
ECTS	6
Arbeitsaufwand	Präsenzstudium: 4 SWS Eigenstudium: 56 h pro Semester, davon • 28 h: Vor- und Nachbereitung der Veranstaltung und freies Selbststudium • 28 h: Vorbereitung für die Prüfung
Vorausgesetzte Kenntnisse	DBSI, Kombinatorik, Statistik
Lehrinhalte	Grundlagen der Anfrageoptimierung
Lern- und Kompetenzziele	Fachkompetenz: • Suchraumgroessen abschätzen • Komplexitätsanalysen • Indexeinsatz (MK1, MK2) Methodenkompetenz: • Systematisches Auszählen und Aufzählen • Analyse komplexer Algorithmen (MF1, MF2, MF3, MF5) Personale Kompetenz: • Präzises Analysieren (MKO1, MKO3)
Medienformen	Präsentationen mit Tafelanschrieb
Begleitende Literatur	Skript
Lehr- und Lernmethoden	Vorlesung (2 SWS)
Art der Prüfungsleistung	Mündliche Prüfung
Prüfungsvorleistung	-
Prüfungsdauer	30 Minuten
Sprache	Deutsch
Angebotsturnus	FSS
Modulverantwortliche/r	Prof. Dr. Guido Moerkotte

Dauer des Moduls	1 Semester
Weiterführende Module	-
Verwendbarkeit	M.Sc. Wirtschaftsinformatik
Einordnung in Fachsemester	1./ 2./ 3. Fachsemester

IE 650: Knowledge Graphs

Module number	IE 650
Title	Knowledge Graphs
Form of module	Lecture
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 124 h per semester, including • 82 h: Pre- and post-lecture studying and revision • 42 h: Examination preparation
Prerequisites	Java or Python programming skills
Aim of module	• The Role of knowledge graphs in the AI landscape • Semantic Web and its representation languages • Labeled property graphs • Query languages for knowledge graphs • Knowledge modeling and ontologies • Logical reasoning with knowledge graphs • Machine learning with knowledge graphs and knowledge graph embeddings
Learning outcomes and qualification goals	Professional competence: • The participants of this course learn about principles and applications of knowledge graphs • They become familiar with their technical foundations such as representation and query languages, or logical inference • After taking this course, the students will be aware of the problems and benefits of knowledge graph technologies in the context of tasks such as knowledge management, information search and data integration • They will be capable of judging the applicability of these technologies for addressing practical challenges (MK1, MK2) Methodological competence: • The participants learn how to design and implement AI systems based on knowledge graphs • They are able to use standardized modeling languages for building knowledge representations and to query these models by means of languages such as SPARQL (MF3) Personal competence: • By jointly building a knowledge graph-based application, the students learn how to effectively work in teams • They improve upon their presentation skills by showing the outcomes of their projects to the other participants of the course (MKO1, MKO3)
Media	Lecture slides and exercise sheets will be available online

Literature	• Pascal Hitzler, Markus Krötzsch and Sebastian Rudolph, <i>Foundations of Semantic Web Technologies</i>, Chapman & Hall/CRC, 2009 • Allemang and Hendler (2008): <i>Semantic Web for the Working Ontologist</i>. Verlag Morgan Kaufmann • Antoniou and van Harmelen (2004): <i>A Semantic Web Primer</i>. MIT Press • Fensel et al. (2020): <i>Knowledge Graphs: Methodology, Tools and Selected Use Cases</i>. Springer • Kerjwal et al. (2021): <i>Knowledge Graphs: Fundamentals, Techniques, and Applications</i>. MIT Press
Methods	The course participants will take part in theoretical and practical exercises, the solutions of which are discussed in the tutorials. At the end of the course, they get the opportunity to apply their knowledge in a team project. Each student team will design and implement a semantic web application, and subsequently present the results to the other students. Besides the exercises, regular presentations including references to relevant course materials and recommended readings will be given by the lecturer. The lecturer as well as the tutors offer individual help and consulting to the participants of the course
Form of assessment	Written examination
Admission requirements for assessment	Project report and oral presentation
Duration of assessment	60 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Heiko Paulheim, Dr. Sven Hertling
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Social Data Science, Teacher Education Informatics
Semester	1st/ 2nd/ 3rd semester

IE 655: Industrial Applications of Artificial Intelligence

Module number	IE 655
Title	Industrial Applications of Artificial Intelligence
Form of module	Lecture and accompanying tutorials
Type of module	Specialization Course
Level	Master
ECTS	9
Workload	Hours in presence: 56 h per semester (2+2 SWS) Self-study: 214 h per semester, including the creation of a learning portfolio
Prerequisites	Necessary Knowledge: • Machine Learning Concepts and Techniques • Programming in Python
Aim of module	Participants will learn about the use of Artificial Intelligence methods, mostly from the field of machine learning in different sectors and industries. They will learn about application areas in the primary, secondary and tertiary sector, get an introduction to examples of such applications that have been published on a scientific level and gather some experience in working with data from the respective fields using publically available datasets
Learning outcomes and qualification goals	Professional competence: • Students will acquire knowledge about possible applications of machine learning in different branches of industry as well as the dominant methods used in these areas (MK2, MK3) Methodological competence: • Identify potential for applying AI methods in different areas of industry • Decide on a suitable method for addressing typical problems in these industries (MF2) Personal competence: • Participants will learn to reflect and document their own learning process (MKO2)
Media	Slides, Data Sets, Software Tools
Literature	Various Scientific Publications - details in the lecture slides
Methods	Lectures, tutorials, independent study
Form of assessment	Learning Portfolio
Admission requirements for assessment	-
Duration of assessment	Continuous assessment throughout the module (learning portfolio)
Language	English

Offering	FSS
Person in charge	Prof. Dr. Heiner Stuckenschmidt
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Social Data Science
Semester	2nd/ 3rd/ 4th semester

IE 670: Web Data Integration

Module number	IE 670
Title	Web Data Integration
Form of module	Lecture
Type of module	Specialization Course
Level	Master
ECTS	3
Workload	Hours in presence: 28 h per semester (2 SWS) Self-study: 56 h per semester, including • 31 h: Pre- and post-lecture studying and revision • 25 h: Examination preparation
Prerequisites	-
Aim of module	Data integration is one of the key challenges in most IT projects and it is estimated that data scientists spend about 80% of their time on data integration and data preparation. Within the enterprise context, data integration problems arise whenever data from separate sources needs to be combined as the basis for new applications or data analysis projects. Within the context of the Web, data integration techniques form the foundation for taking advantage of the ever-growing number of publicly-accessible data sources. The course will cover the following topics: • Heterogeneity and Distributedness • The Data Integration Process • Structured Data on the Web • Data Exchange Formats • Schema Mapping and Data Translation • Identity Resolution • Data Quality Assessment • Data Fusion It is highly recommended to attend the course IE683 Web Data Integration Project in the same semester as this course as the schedules of both courses are aligned to each other
Learning outcomes and qualification goals	Professional competence: • Students will be able to identify opportunities for employing Web data in business applications and will learn to select and apply appropriate techniques for integrating and cleansing Web data (MK1, MF1) Methodological competence: • Participants will acquire knowledge of the data integration process as well as the techniques that are used in each phase of the process (MK2, MF3, MF4, MKO3) Personal competence: -
Media	Slide set

Literature	• AnHai Doan, Alon Halevy, Zachary Ives: <i>Principles of Data Integration</i>. Morgan Kaufmann, 2012 • Luna Dong, Divesh Srivastava: <i>Big Data Integration</i>. Morgan & Claypool, 2015 • Ulf Leser, Felix Naumann: <i>Informationsintegration</i>. Dpunkt Verlag, 2007
Methods	The course consists of a lecture that introduces students to state of the art data integration techniques
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	60 minutes
Language	English
Offering	HWS
Person in charge	Dr. Ralph Peeters
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Social Data Science, Teacher Education Informatics
Semester	1st/ 2nd/ 3rd semester

IE 675b: Machine Learning

Module number	IE 675b
Title	Machine Learning
Form of module	Lecture with exercise
Type of module	Fundamental course (MMDS), Specialization Course (M.Sc. Wifo)
Level	Master
ECTS	9
Workload	Hours in presence: 63 h per semester (4+2 SWS) Exercises, assignments, and self-study: 167 h per semester
Prerequisites	IE 500 Data Mining I (recommended), probability/linear algebra
Aim of module	Machine learning is concerned with building computer systems that improve with experience as well as the study of learning processes, including the design of algorithms that are able to make predictions or extract knowledge from data. The aim of this module is to provide an introduction into the field of machine learning, and study algorithms, underlying concepts, and theoretical principles. Topics include: • Basics of machine learning • Estimators, inference (MLE, ERM, RRM as well as MAP/Bayesian inference) and decision • Selected classification and regression models • Parameter estimation (MLE, MAP, ERM, RRM, Bayesian, first- and second-order methods) • Dimensionality reduction (SVD, PCA, FA) • Expectation maximization and mixture models • Kernels, kernel trick, vector machines and sparsity • Hyperparameter optimization
Learning outcomes and qualification goals	Professional competence: • Deep understanding of algorithms and underlying concepts of machine learning (MK1, MF1) Methodological competence: • Being able to apply machine learning techniques and systems for a given problem • Being able to model and implement new machine learning techniques (MK2, MF3, MF4) Personal competence: • Writing skills • Presentation skills • Statistical programming skills (MKO3, MF2)
Media	Slide set, exercise sheets, software, datasets
Literature	• K.P. Murphy. <i>Probabilistic Machine Learning: An Introduction</i>. The MIT Press, 2022 • Additional material, books and articles provided in lecture notes

Methods	The course consists of a lecture accompanied by theoretical and practical exercises as well as case studies with real data. In the exercises, students will deepen the material discussed in the lecture, apply the methods in practice and present the result
Form of assessment	Written examination
Admission requirements for assessment	Homework assignments (pass at least 3 assignments)
Duration of assessment	120 minutes
Language	English
Offering	HWS
Person in charge	Prof. Dr. Rainer Gemulla
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	1st/ 2nd/ 3rd semester

IE 678: Deep Learning

Module number	IE 678
Title	Deep Learning
Form of module	Lecture with exercise
Type of module	Data analytics methods (MMDS), Specialization Course (M.Sc. Wifo)
Level	Master
ECTS	6
Workload	Hours in presence: 42 h per semester (2+2 SWS) Exercises, assignments and self-study: 119 h per semester
Prerequisites	IE 675b Machine Learning or equivalent
Aim of module	Building upon IE 675b Machine Learning, this course focuses on deep learning and introduces basic and advanced deep learning concepts, architectures, training methods as well as selected applications. Topics include: • Feedforward neural networks • Backpropagation, optimizers and architecture design • Machine learning systems • Local and global embeddings • Convolutional neural networks • Recurrent neural networks and structured state space models • Attention and Transformers • Deep learning for graphs • Training techniques for deep learning models, including data augmentation, pretraining and fine-tuning and in-context learning
Learning outcomes and qualification goals	Professional competence: • Deep understanding of fundamental concepts, models and algorithms of deep learning (MK1, MF1) Methodological competence: • Being able to build and train deep learning models • Being able to select suitable deep learning techniques for a given learning problem (MK2, MF3, MF4) Personal competence: • Writing skills • Presentation skills • Statistical programming skills (MKO3, MF2)
Media	Slide set, exercise sheets, software, datasets

Literature	• K.P. Murphy. <i>Probabilistic Machine Learning: An Introduction</i>. The MIT Press, 2022 • K.P. Murphy. <i>Probabilistic Machine Learning: Advanced Topics</i>. The MIT Press, 2023 • I. Goodfellow, Y. Bengio, A. Courville. <i>Deep Learning</i>. The MIT Press, 2017 • I. Drori. <i>The Science of Deep Learning</i>. Cambridge University Press, 2023 • Additional material and articles provided in lecture notes
Methods	The course consists of a lecture accompanied by theoretical and practical exercises as well as case studies with real data. In the exercises, students will deepen the material discussed in the lecture, apply the methods in practice, and present the result
Form of assessment	Oral examination
Admission requirements for assessment	Homework assignments (pass at least 2 assignments)
Duration of assessment	25 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr. Rainer Gemulla
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	2nd/ 3rd semester

IE 679: AI Deployment and Operations

Modulnummer	IE 679
Titel	AI Deployment and Operations
Form der Veranstaltung	Online-Vorlesung + praktische Gruppenarbeit
Typ der Veranstaltung	Spezialisierung / Data Analysis Methods
Modulniveau	Master
ECTS	3
Arbeitsaufwand	Präsenzstudium: 14 h pro Semester (1 SWS) Eigenstudium: 75 h pro Semester, davon • 60 h: Vor- und Nachbereitung der Veranstaltung und freies Selbststudium • 15 h: Vorbereitung für die Prüfung
Vorausgesetzte Kenntnisse	Machine Learning oder Data Mining I
Lehrinhalte	• Grundsätze von Cloudcomputing und praktische Anwendung • Infrastructure as Code • Gängige Infrastrukturen • Erstellung von replizierbarem Code nach Industriestandards • Erstellung von Dashboards zur Analyse von Daten
Lern- und Kompetenzziele	Fachkompetenz: • Unterschiede zwischen akademischen und industriellen KI-Anwendungen • Kenntnisse über den Infrastruktur-Stack für industrielle KI-Anwendungen • Voraussetzungen und Methoden für die Realisierbarkeit von Code Methodenkompetenz: • Erstellung von praxistauglichen KI Anwendungen Personale Kompetenz: • Besseres Verständnis der Anforderungen an nicht akademische Tätigkeiten
Medienformen	Präsentation
Begleitende Literatur	• Cloud Computing Buch: Learning Microsoft Azure: <i>Cloud Computing and Development Fundamentals</i> von Jonah Carrio Andersson • Dokumentation MS Azure: https://learn.microsoft.com/en-us/training/azure/ • Dokumentation Terraform: https://developer.hashicorp.com/terraform/intro
Lehr- und Lernmethoden	• Vorlesung (2 SWS) jede 2. Woche • Praktische Arbeit in Kleingruppen
Art der Prüfungsleistung	Abgabe eines Codingprojekts + Gruppenpräsentation

Prüfungsvorleistung	Dokumentation der praktischen Arbeiten
Prüfungsdauer	10-15 Minuten pro Teilnehmer
Sprache	Englisch
Angebotsturnus	HWS
Modulverantwortliche/r	Prof. Dr. Heiner Stuckenschmidt
Dauer des Moduls	1 Semester
Weiterführende Module	-
Verwendbarkeit	M.Sc. Wirtschaftsinformatik, M.Sc. Data Science, M.Sc. Social Data Science
Einordnung in Fachsemester	2./ 3./ 4. Fachsemester

IE 683: Web Data Integration Project

Module number	IE 683
Title	Web Data Integration Project
Form of module	Project
Type of module	Specialization Course
Level	Master
ECTS	3
Workload	Hours in presence: 28 h per semester (2 SWS) Self-study: 56 h per semester, including • 36 h: Project work • 20 h: Report writing and presentation preparation
Prerequisites	Programming skills in Python
Aim of module	The web data integration project allows students to apply the methods and techniques that they have learned in the lecture Web Data Integration in the context of a practical integration project. The projects cover all steps of the data integration process including data gathering, schema mapping, data translation, identity resolution, data quality assessment, and data fusion. It is highly recommended to attend the web data integration lecture (IE670) in the same semester as the web data integration project as the schedules of both courses are aligned to each other
Learning outcomes and qualification goals	Professional competence: • Students will be able to identify opportunities for employing Web data in business applications and will learn to apply appropriate techniques for integrating and cleansing Web data (MK1, MF1) Methodological competence: • Participants will acquire knowledge of the data integration process as well as the techniques that are used in each phase of the process • Project organization skills (MK2, MF3, MF4, MKO3) Personal competence: • Presentation skills • Team work skills (MKO2, MF2)
Media	Exercise sheets, Java project template
Literature	• AnHai Doan, Alon Halevy, Zachary Ives: <i>Principles of Data Integration</i>. Morgan Kaufmann, 2012 • Luna Dong, Divesh Srivastava: <i>Big Data Integration</i>. Morgan & Claypool, 2015 • Ulf Leser, Felix Naumann: <i>Informationsintegration</i>. Dpunkt Verlag, 2007
Methods	Students work on their integration projects in teams and will report about the results of their projects in the form of a written report as well as an oral presentation

Form of assessment	• Project report (70%, 10-20 pages in length) • Oral project presentation (30%, 15-20 minutes)
Admission requirements for assessment	-
Duration of assessment	-
Language	English
Offering	HWS
Person in charge	Dr. Ralph Peeters
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Social Data Science, Teacher Education Informatics
Semester	1st/ 2nd/ 3rd semester

IE 685: Large Language Models and Agents

Module number	IE 685
Title	Large Language Models and Agents
Form of module	Lecture
Type of module	Specialization Course
Level	Master
ECTS	3
Workload	Hours in presence: 28 h per semester (2 SWS) Self-study: 56 h per semester, including • 31 h: Pre and post lecture studying and revision • 25 h: Examination preparation
Prerequisites	• Machine learning concepts and techniques • Knowledge about natural language processing
Aim of module	Large Language Models (LLMs) such as GPT, Llama, Gemini and Claude, have the potential to enable a wide range of new applications and to significantly improve the performance of existing systems. The course introduces students to LLMs as well as their application. The course covers the following topics: • Introduction to LLMs • Prompt engineering • LLM-based agents • Tool use and environment interaction • Retrieval augmented generation • Context engineering for LLM agents • Safety and security of LLM agents • Evaluation of LLM agents It is highly recommended to attend the course IE 686 Large Language Models and Agents Project in the same semester as this course as the schedules of both courses are aligned to each other
Learning outcomes and qualification goals	Professional competence: • Students will be able to identify opportunities for employing Large Language Models in business applications and will learn to select and apply appropriate prompt/context engineering techniques for solving complex tasks (MK1, MF1) Methodological competence: • Participants will acquire knowledge on the application of Large Language Models and LLM agents for solving real-world problems (MK2, MF3, MF4, MKO3) Personal competence: -
Media	Slide sets

Literature	• Zhao, et al.: <i>A Survey of Large Language Models</i>. arXiv:2303.18223v16, 2025 • Wang, et al.: <i>A Survey on Large Language Model based Autonomous Agents</i>. Frontiers of Computer Science, 2024 • Mohammadi, et al. <i>Evaluation and benchmarking of LLM agents: A survey</i>. SIGKDD Conference on Knowledge Discovery and Data Mining, 2025
Methods	Methods The course consists of a lecture that introduces students to Large Language Models and state of the art prompt/context engineering techniques
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	60 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr. Christian Bizer
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science, M.Sc. Social Data Science, M.Sc. Business Informatics, Teacher Education Informatics
Semester	2nd/ 3rd semester

IE 686: Large Language Models and Agents Project

Module number	IE 686
Title	Large Language Models and Agents Project
Form of module	Project
Type of module	Specialization Course
Level	Master
ECTS	3
Workload	Hours in presence: 28 h per semester (2 SWS) Self-study: 56 h per semester, including • 36 h: Project work • 20 h: Report writing and presentation preparation
Prerequisites	• Programming skills in Python • Machine learning concepts and techniques
Aim of module	The Large Language Models and Agents project allows students to apply the methods and techniques that they have learned in the lecture Large Language Models and Agents (IE 685) in the context of a practical project. The projects cover all steps of the agent development cycle including agentic workflow design, prompt- and context engineering, and agent evaluation. It is highly recommended to attend the Large Language Models and Agents lecture (IE 685) in the same semester as the Large Language Models and Agents project
Learning outcomes and qualification goals	Professional competence: • Students will be able to identify opportunities for employing large language models and agents in business applications and will learn to select and apply appropriate prompt/context engineering techniques for solving complex tasks (MK1, MF1) Methodological competence: • Participants will acquire knowledge on the application of large language models and related techniques for solving complex real-world problems • Agent development using LangChain framework • Project organization skills (MK2, MF3, MF4, MKO3) Personal competence: • Presentation skills • Teamwork skills • Python programming skills (MKO2, MF2)
Media	Exercise sheets, software, datasets

Literature	• Zhao, et al.: <i>A Survey of Large Language Models</i>. arXiv:2303.18223v16, 2025 • Wang, et al.: <i>A Survey on Large Language Model based Autonomous Agents</i>. Frontiers of Computer Science, 2024 • Mohammadi, et al. <i>Evaluation and benchmarking of LLM agents: A survey</i>. SIGKDD Conference on Knowledge Discovery and Data Mining, 2025
Methods	Students work on their projects in teams and will report about the results of their projects in the form of a written report as well as an oral presentation
Form of assessment	• Project report (50%, 10-20 pages in length) • Oral project presentation and Q&A (50%, 15-20 minutes)
Admission requirements for assessment	-
Duration of assessment	-
Language	English
Offering	FSS
Person in charge	Prof. Dr. Christian Bizer
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science, M.Sc. Social Data Science, M.Sc. Business Informatics, Teacher Education Informatics
Semester	2nd/ 3rd semester

IE 692: Advanced Process Mining

Module number	IE 692
Title	Advanced Process Mining
Form of module	Lecture with exercise
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	150 h per semester
Prerequisites	Familiarity with Petri nets and BPMN for process modeling, along with basic Python programming skills, is expected. Prior experience in process mining is not a prerequisite
Aim of module	Process mining is a data-driven discipline that extracts qualitative and quantitative insights into the execution of organizational processes through the analysis of event data. It combines formal process models with data science and machine-learning techniques. The course covers the foundations, algorithms, and state-of-the-art methods of process mining, including • Process discovery • Conformance checking • Process enhancement • Predictive monitoring • Non-traditional event data, including object-centric and partially ordered event data • Event data preprocessing, abstraction, and clustering Lectures are complemented by hands-on exercises that primarily focus on pen-and-paper problem solving, supported by selected implementations and evaluations using open-source process mining tools. A group project allows students to apply and evaluate process mining and ML-based methods in a practical setting
Learning outcomes and qualification goals	Knowledge: • After completing the course, students will be familiar with both fundamental and state-of-the-art process mining techniques and algorithms for the data-driven analysis of business processes (MK1, MK2) Capabilities: • After completing the course, students will be able to analyze business processes using process mining techniques and (open-source) process mining tools (MF1, MF2, MF3) Competencies: • During the course, students will learn to apply, compare, and evaluate process mining techniques on synthetic and real-world data, both individually and in teams (MKO1, MKO3)
Media	Electronic slides and exercise sheets

Literature	• van der Aalst: <i>Process Mining: Data Science in Action</i>, 2nd edition, Springer (2016) DOI: 10.1007/978-3-662-49851-4 • van der Aalst & Carmona (Editors): <i>Process Mining Handbook</i>, Springer (2022) DOI: 10.1007/978-3-031-08848-3 • Carmona et al.: <i>Conformance Checking: Relating Processes and Models</i>, Springer (2018) DOI: 10.1007/978-3-319-99414-7 • Additional academic papers will be recommended on a per-lecture basis
Methods	• Lectures • Homework exercises to practice with joint discussion in exercise sessions
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr. Daniel Schuster
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Social Data Science
Semester	2nd semester

IE 695: Reinforcement Learning

Module number	IE 695
Title	Reinforcement Learning
Form of module	Lecture with exercise (partially online)
Type of module	Data Analytics Methods
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 98 h per semester, including • 70 h: Lectures/exercises • 28 h: Examination preparation
Prerequisites	Machine Learning / Computer Vision / Generative Computer Vision Models course, theoretical and practical knowledge of neural networks
Aim of module	• Basic concepts of reinforcement learning: MDP, policies, on-policy, off-policy learning • Classical tabular reinforcement learning, DP, Policy Iteration, Q-Learning, SARSA, Monte-Carlo methods • Function approximation for reinforcement learning • Policy gradient methods
Learning outcomes and qualification goals	Professional competence: • After the course students will understand classical concepts of reinforcement learning as well as state-of-the-art algorithms Methodological competence: • Understand and customize popular reinforcement learning algorithms • Choose the right setting for their problem • Train agents to perform well in environments with which they interact Personal competence: • Trains abstract thinking and formal modeling of application scenarios • By solving assigned exercises independently, the transfer of the learned material to related questions is promoted
Media	Exercise sheets and lecture slides available online, blackboard
Literature	Sutton & Barto: <i>Reinforcement Learning: An Introduction</i> , 2018
Methods	Lecture, exercises every two weeks, book studies
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	HWS

Person in charge	Prof. Dr.-Ing. Margret Keuper
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	1st/ 2nd/ 3rd semester

IE 696: Advanced Methods in Text Analytics

Module number	IE 696
Title	Advanced Methods in Text Analytics
Form of module	Lecture with exercise
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester (4 SWS) Self-study: 112 h per semester, including • 84 h: Pre- and post-lecture studying and revision • 28 h: Examination preparation
Prerequisites	• Fundamental notions of linear algebra and probability theory • Successful completion of “Text Analytics” (IE661), “Machine Learning” (IE675b) or “Deep Learning” (IE678)
Aim of module	This module builds upon the introduction to Natural Language Processing (NLP) from “Text Analytics” (IE661) and introduces students to cutting-edge problems, techniques, and state-of-the-art methods in NLP. The course will focus on neural models of meaning in context and present a variety of “Deep Learning” architectures for different applications in human language technology (e.g., summarization, dialogue systems, and machine translation). Moreover, we will cover open research areas, such as the explainability and interpretability of NLP models and methods to quantify the degree of bias they exhibit
Learning outcomes and qualification goals	Professional competence: • Students will acquire knowledge of state-of-the-art principles and methods of Natural Language Processing, specifically focusing on applying statistical methods to human language technologies (MK1, MK2, MF3) Methodological competence: • Successful participants will be able to understand state-of-the-art methods for Natural Language Processing and select, apply and evaluate the most appropriate techniques for various practical and application-oriented scenarios (MF3) Personal competence: -
Media	Lecture and tutorial slides, exercise sheets
Literature	Dan Jurafsky and James H. Martin: <i>Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition</i> (3rd edition, online: https://web.stanford.edu/~jurafsky/slp3/)
Methods	Lectures, tutorials
Form of assessment	Written examination

Admission requirements for assessment	-
Duration of assessment	90 minutes
Language	English
Offering	FSS
Person in charge	Prof. Dr. Simone Paolo Ponzetto
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Social Data Science, Teacher Education Informatics
Semester	1st/ 2nd/ 3rd semester

IE 698: Foundations and Applications of Digital Health Technologies

Module number	IE 698
Title	Foundations and Applications of Digital Health Technologies
Form of module	Lecture
Type of module	Specialization Course
Level	Master
ECTS	3
Workload	Hours in presence: 28 h per semester Self-study: 56 h per semester
Prerequisites	Necessary: • Basic knowledge about AI systems (knowledge-based systems, machine learning, deep neural networks) Optional: • Participation of Lecture “Responsible AI” Winter Term 2023/24 • Basic knowledge about digital health systems
Aim of module	Conceptual foundations: • Understanding of important concepts in digital health (eHealth, mHealth, telehealth, digital phenotyping, digital twins, and other concepts) Methods: • e.g., interviewing, group discussions, design research methods (prototyping, design thinking, co-design), (digital) ethnography Applications: • AI in digital health apps in medicine and healthcare • AI-based robots • AI-based VR systems
Learning outcomes and qualification goals	Knowledge: • Students gain insights and understanding of important concepts in digital health • They learn modes of transdisciplinary thinking and theorizing on digital health (MK1) Capabilities: • Students learn elements of mixed-methods study design for digital health research and co-design methodology • They learn to assess scientific publications in this domain critically (MF1, MF2) Competencies: • Students learn to critically assess the conceptual, technical, ethical, legal and social aspects of digital health applications (MKO1)
Media	Slides and supporting media will be available online

Literature	• Fagherazzi, G. (2020). <i>Deep Digital Phenotyping and Digital Twins for Precision Health: Time to Dig Deeper</i>. Journal of Medical Internet Research, 22, e16770. https://www.jmir.org/2020/3/e16770/ • Hahn, H. & Schreiber, A. (2019). <i>E-Health</i>. In Neugebauer, R. (Ed.), <i>Digital Transformation</i> (pp. 311-334). Springer. https://doi.org/10.1007/978-3-662-58134-6_19 • Budd, J. et al. (2020). <i>Digital technologies in the public-health response to COVID-19</i>. Nat Med, 26, 1183-1192. https://www.nature.com/articles/s41591-020-1011-4 • Voeneky, S., Kellmeyer, P., Mueller, O., & Burgard, W. (Eds.). (2022). <i>The Cambridge Handbook of Responsible Artificial Intelligence: Interdisciplinary Perspectives</i>. Cambridge University Press. https://doi.org/10.1017/9781009207898
Methods	Interactive lecture
Form of assessment	Written examination
Admission requirements for assessment	-
Duration of assessment	45 minutes
Language	English
Offering	FSS
Person in charge	Jun.-Prof. Dr. Philipp Kellmeyer
Duration of module	1 semester
Further modules	Lecture on Responsible AI, Fall 2024
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	All semesters possible

IE 699: Co-creating digital health applications with design methodology

Module number	IE 699
Title	Co-creating digital health applications with design methodology
Form of module	Exercise
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 56 h per semester Self-study: 112 h per semester
Prerequisites	Necessary: • Basic knowledge about AI systems (knowledge-based systems, machine learning, deep neural networks) Optional: • Participation of Lecture “Responsible AI” Winter Term 2023/24 • Basic knowledge about digital health systems
Aim of module	Conceptual foundations: • Understanding of important concepts in digital health (eHealth, mHealth, telehealth, digital phenotyping, digital twins and other concepts) Methods: • Interviewing, group discussions, design research methods (prototyping, design thinking, co-design), (digital) ethnography Applications: • By-design approaches to developing digital health apps and other solutions
Learning outcomes and qualification goals	Knowledge: • Along with digital health use cases, they learn basic by-design principles for developing digital health applications (MK1) Capabilities: • Students learn elements of mixed-methods study design for digital health research and co-design methodology via design fictions and mockups developed in groups in the exercise (MF1, MF2, MF3) Competencies: • Students learn to apply their knowledge and capabilities regarding by-design approaches to solve specific problems in a team context (MKO1)
Media	Slides and supporting media will be available online

Literature	• Fagherazzi, G. (2020). <i>Deep Digital Phenotyping and Digital Twins for Precision Health: Time to Dig Deeper</i>. Journal of Medical Internet Research, 22, e16770. https://www.jmir.org/2020/3/e16770/ • Hahn, H. & Schreiber, A. (2019). <i>E-Health</i>. In Neugebauer, R. (Ed.), <i>Digital Transformation</i> (pp. 311-334). Springer. https://doi.org/10.1007/978-3-662-58134-6_19 • Budd, J. et al. (2020). <i>Digital technologies in the public-health response to COVID-19</i>. Nat Med, 26, 1183-1192. https://www.nature.com/articles/s41591-020-1011-4 • Voeneky, S., Kellmeyer, P., Mueller, O., & Burgard, W. (Eds.). (2022). <i>The Cambridge Handbook of Responsible Artificial Intelligence: Interdisciplinary Perspectives</i>. Cambridge University Press. https://doi.org/10.1017/9781009207898
Methods	Exercise
Form of assessment	Presentation in course
Admission requirements for assessment	-
Duration of assessment	20 minutes (15 minutes Presentation, 5 minutes Q&A)
Language	English
Offering	FSS
Person in charge	Jun.-Prof. Dr. Philipp Kellmeyer
Duration of module	1 semester
Further modules	Lecture on Responsible AI, Fall 2024
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	All semesters possible

IS-Courses:

Module no.	Name of Module	Offered	Language	ECTS	Page
IS 512 (IS 5120)	IT Management in the Digital Age	MMM*	E	6	MMM*
IS 513 (IS 5130)	Applied IT Management in the Digital Age	MMM*	E	6	MMM*
IS 540 (IS 5400)	Management of Enterprise Systems	MMM*	E	6	MMM*
IS 541 (IS 5410)	Methods and Theories in Information Systems (ManTIS)	MMM*	E	6	MMM*
IS 603 (IS 6030)	IT Leadership in Organizations	MMM*	E	6	MMM*
IS 607 (IS 6070)	Digital Innovation	MMM*	E	6	MMM*
IS 609 (IS 6090)	AI and Technology Strategy	MMM*	E	6	MMM*
IS 611 (IS 6110)	Advanced Topics in Large Language Models	MMM*	E	6	MMM*
IS 612 (IS 6120)	Product Experimentation and Analytics	MMM*	E	6	MMM*
IS 613 (IS 6130)	Applied Project in Design Thinking and Lean Software Development	MMM*	E	6	MMM*
IS 614 (IS 6140)	Corporate Knowledge Management	MMM*	E	6	MMM*
IS 615 (IS 6150)	Enterprise Cloud Design and Development	MMM*	E	6	MMM*
IS 617 (IS 6170)	Large Language Models for the Economic and Social Sciences	MMM*	E	6	MMM*
IS 629 (IS 6290)	Agile Software Product Management and Design	MMM*	E	6	MMM*
IS 661 (IS 6610)	Text Analytics	MMM*	E	6	MMM*

* For a detailed description, please see the module catalogue of the respective following degree programs:

MMM: M.Sc. Mannheim Master in Management, <https://www.bwl.uni-mannheim.de/en/module-catalog-mmm/>

** Prerequisite: No completed exam in IE 676

International Courses and other Specialization Courses:

BI 656: International Course

Module number	BI 656
Title	International Course
Form of module	Depends on course taken abroad
Type of module	Specialization Course
Level	Master
ECTS	Max. 18
Workload	Depends on course taken abroad
Prerequisites	Depends on course taken abroad
Aim of module	The course level equals a regular 600-level course in the MSc. Business Informatics program. The module can be taken during a study abroad term / semester and complements the Mannheim curriculum of the student
Learning outcomes and qualification goals	Depends on course taken abroad
Media	Depends on course taken abroad
Literature	Depends on course taken abroad
Methods	Depends on course taken abroad
Form of assessment	Depends on course taken abroad
Admission requirements for assessment	-
Duration of assessment	Depends on course taken abroad
Language	English preferred, but any other language possible if Mannheim faculty member is able to identify content and level
Offering	HWS/FSS
Person in charge	Lecturer at the host university
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics
Semester	2nd/ 3rd/ 4th semester

DS 203: Responsible AI: Conceptual Foundations, Methods and Applications

Module number	DS 203
Title	Responsible AI: Conceptual Foundations, Methods and Applications
Form of module	Lecture with Essay
Type of module	Specialization Course
Level	Master
ECTS	6
Workload	Hours in presence: 28 h per semester (2 SWS) Self-study: 56 h lectures; 20 h essay / preparation oral exam
Prerequisites	Basic knowledge about AI systems (knowledge-based systems, machine learning, deep neural networks)
Aim of module	Conceptual foundations: • Understanding of important concepts in human-AI interaction and AI ethics (such as trust, autonomy, responsibility) Methods: • e.g., narrative interviews, group discussions, design research methods (prototyping, design thinking, techno-mimesis), (digital) ethnography, participatory action research Applications: • AI in medicine and healthcare • Social robotics • Generative AI • Other use cases / real-world AI applications
Learning outcomes and qualification goals	Professional competence: • Students gain insights and understanding of important concepts in human-AI interaction and AI ethics • They learn modes of transdisciplinary thinking and theorizing • Along sector-specific use cases they learn about ethical, legal and social aspects and challenges of real-world AI application, e.g. for healthcare Methodological competence: • Students learn elements of mixed-methods study design for human-AI interaction research Personal competence: • Students learn to critically assess conceptual, ethical, legal and social aspects of human-AI interaction • They gain skills in transdisciplinary research and theory-building and learn to transfer these insights to real-world human-AI interaction scenarios
Media	Slides are available online

Literature	• Voeneky, S., P. Kellmeyer, O. Mueller, and W. Burgard, ed. (2022). <i>The Cambridge Handbook of Responsible Artificial Intelligence: Interdisciplinary Perspectives</i>. Cambridge University Press. https://doi.org/10.1017/9781009207898 • Coeckelbergh, Mark. (2020). <i>AI Ethics</i>. The MIT Press. https://www.gbv.de/dms/bowker/toc/9780262538190.pdf • Heilinger, J.-C. (2022). <i>The Ethics of AI Ethics. A Constructive Critique</i>. <i>Philosophy & Technology</i>, 35(3), 61. https://doi.org/10.1007/s13347-022-00557-9 • McLennan, S., Fiske, A., Tigard, D., Müller, R., Haddadin, S., & Buyx, A. (2022). <i>Embedded ethics: A proposal for integrating ethics into the development of medical AI</i>. <i>BMC Medical Ethics</i>, 23(1), 6. https://doi.org/10.1186/s12910-022-00746-3 • Schmitt, L. (2021). <i>Mapping global AI governance: A nascent regime in a fragmented landscape</i>. <i>AI and Ethics</i>. https://doi.org/10.1007/s43681-021-00083-y
Methods	Interactive lecture
Form of assessment	Essay
Admission requirements for assessment	-
Duration of assessment	Essays need to be handed in by December 8th
Language	English
Offering	HWS
Person in charge	Jun.-Prof. Dr. Philipp Kellmeyer
Duration of module	1 semester
Further modules	Follow-up (block) seminar planned for summer semester
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	All semesters possible

MAB 519: Reinforcement Learning

Module number	MAB 519
Title	Reinforcement Learning
Form of module	Lecture with Exercises
Type of module	Mathematics elective B
Level	Master
ECTS	10
Workload	Lectures: 56 h per semester Exercise classes: 28 h per semester Self-study: 156 h per semester
Prerequisites	Stochastics I, Markov Chains
Aim of module	• Foundations of Markov Decision Processes • Policy- and Value-Iteration Methods • Temporal Difference Learning • Policy-Gradient Methods • Implementation in Python
Learning outcomes and qualification goals	• MK1, M02, M03 • MF1, MF3 • (cf. “Erläuterungen zu den Abkürzungen”)
Media	Blackboard, Slides
Literature	• Lecture notes • Sutton, Barto: <i>Reinforcement Learning - An Introduction</i> • Putterman: <i>Markov Decision Processes</i>
Methods	Lectures, theoretical and programming exercises
Form of assessment	Oral examination
Admission requirements for assessment	Participation in the exercises
Duration of assessment	30 minutes
Language	English
Offering	Irregular
Person in charge	Prof. Dr. Leif Döring
Duration of module	1 semester
Further modules	-

Programs	M.Sc. Mathematics in Business and Economics, B.Sc. Mathematics in Business and Economics, M.Sc. Mathematics, M.Sc. Data Science, M.Sc. Business Informatics, Teacher Education Mathematics, M.Sc. Economic and Business Education
Semester	1st/ 2nd/ 3rd semester

MAC 404: Lineare Optimierung

Modulnummer	MAC 404
Titel	Lineare Optimierung / <i>Linear Optimization</i>
Form der Veranstaltung	Vorlesung mit Übung
Typ der Veranstaltung	Wahlpflichtveranstaltung Mathematik C
Modulniveau	Bachelor
ECTS	8
Arbeitsaufwand	Präsenzstudium: 84 h pro Semester (6 SWS) Eigenstudium: 154 h pro Semester, davon • 126 h: Vor- und Nachbereitung der Veranstaltung und freies Selbststudium • 28 h: Vorbereitung für die Prüfung
Vorausgesetzte Kenntnisse	Lineare Algebra I & II/A, Analysis I & II
Lehrinhalte	• Geometrie von Polyedern • Theorie der Linearen Optimierung • Ganzzahlige Optimierung • Innere-Punkt-Verfahren • Optimierung auf Graphen
Lern- und Kompetenzziele	Fachkompetenz: • Verständnis der wesentlichen Konzepte und Lösungsverfahren der Linearen Optimierung (BF1, BK1) • Computerunterstützte Umsetzung anwendungsbezogener Fragestellungen (BK2, BK3, BO1) • Querverbindungen zu anderen mathematischen Gebieten identifizierten Klassifikation und Interpretation numerischer Probleme (BK1, BO2) Methodenkompetenz: • Mathematische Modellierung eines Problems (BF3, BO3) • Konkrete Problemlösungsstrategien und deren Interpretation (BF1, BF2) Personale Kompetenz: • Teamarbeit (BO1, BF4, BF5)
Medienformen	Präsentationen mit Tafelanschriften, eigenes Skriptum
Begleitende Literatur	• Robert J. Vanderbei: <i>Linear Programming: Foundations and Extensions</i> • Alexander Schrijver: <i>Theory of Linear and Integer Programming</i> • Michele Conforti, Gerard Cornuéjols, Giacomo Zambelli: <i>Integer Programming</i>
Lehr- und Lernmethoden	Vorlesung (4 SWS), Übung (2 SWS)

Art der Prüfungsleistung	Schriftliche Prüfung
Prüfungsvorleistung	Bearbeitung von Übungsblättern und mindestens 50 % der Übungsaufgaben bestanden
Prüfungsdauer	90 Minuten
Sprache	Deutsch, auf Wunsch auch Englisch
Angebotsturnus	HWS
Modulverantwortliche/r	Prof. Dr. Mathias Staudigl
Dauer des Moduls	1 Semester
Weiterführende Module	Nonlinear Optimization, Konvexe Optimierung, Optimale Kontrolle
Verwendbarkeit	B.Sc. Wirtschaftsmathematik, M.Sc. Wirtschaftsmathematik, M.Sc. Wirtschaftspädagogik, Lehramt Mathematik, B.Sc. Wirtschaftsinformatik, M.Sc. Wirtschaftsinformatik
Einordnung in Fachsemester	5./ 6. Fachsemester

D. Projects and Seminars

1. Overview

Team Project:

Module no.	Name of Module	Offered	Language	ECTS	Page
TP 500	Team Project	HWS/FSS	E	12	78

Scientific Research:

Module no.	Name of Module	Offered	Language	ECTS	Page
SQ 500	Scientific Research	HWS/FSS	E	2	80

Seminars:

Module no.	Name of Module	Offered	Language	ECTS	Page
CS 701	Selected Topics in Algorithmics and Cryptography	Irregular	E	4	82
CS 704	Master Seminar Artificial Intelligence	Irregular	E	4	84
CS 707	Data Analytics Seminar	Irregular	E	4	86
CS 708	Seminar Software Engineering	Irregular	E	4	88
CS 709	Seminar Text Analytics	Irregular	G/E	4	90
CS 710	Selected Topics in Data Science	Irregular	G/E	4	92
CS 715	Large-Scale Data Integration Seminar	Irregular	E	4	94
CS 716	IT-Security	Irregular	E	4	96
CS 717	Master Seminar on Computer Vision	HWS/FSS	E	4	98
CS 718	AI and Data Science in Fiction and Society	HWS	E	4	100
CS 720	Uncertainty Estimation	Irregular	E	4	102
CS 721	Seminar Data-Science I	Irregular	E	4	104
CS 730	Advanced Implementation Techniques for Database Systems	Irregular	E	4	106
CS 731	Database Theory	Irregular	E	4	108
CS 733	Advanced Topics in Process Mining	FSS	E	4	110
IS 703 (IS 7030)	Master Seminar AI, Platforms, and the Digital Economy	MMM*	E	4	MMM*

IS 712 (IS 7120)	Contemporary Issues in Information Systems Research	Irregular	E	4	112
IS 742 (IS 7420)	Seminar Trends in Enterprise Systems	MMM*	E	4	MMM*
IS 751 (IS 7510)	E-Government Adoption and Societal Change	MMM*	E	4	MMM*

* For a detailed description, please see the module catalogue of the respective following degree programs:

MMM: M.Sc. Mannheim Master in Management: <https://www.bwl.uni-mannheim.de/en/module-catalog-mmm/>

2. Detailed descriptions

TP 500: Team Project

Module number	TP 500
Title	Team Project
Form of module	Project
Type of module	Team Project
Level	Master
ECTS	12 in two consecutive semesters (12 month) or 12 in one semester (6 month)
Workload	Hours in presence: • 12 month-project: 28 h per semester (2 SWS) • 6 month-project: 56 h per semester (4 SWS) Self-study (12 month project): 140 h per semester • 112 h: Pre- and post-lecture studying, revision and free self-study • 28 h: Preparation of examination/presentation Self-study (6 month project): 280 h per semester • 224 h: Pre- and post-lecture studying, revision and free self-study • 56 h: Preparation of examination/presentation
Prerequisites	Depends on topic
Aim of module	The students solve a practical problem as a team. The participants have to analyze and refine the problem and come up with a project plan for developing a concrete solution that will be carried out by the team over the duration of a whole year. Concrete topics for projects are defined by the supervisors and offered to the students who can apply for different topics. Problem area and techniques involved depend on the expertise of the offering chair
Learning outcomes and qualification goals	Professional competence: Depending on the actual topic of the project, participants will acquire: • In-depth knowledge in a certain application of business informatics and data science • Knowledge about methods and technologies typically applied in the application area • Knowledge about practical problems and challenges when applying a certain technique in a given application area (MK1, MK2, MK3) Methodological competence: Participants will learn to: • Refine a given problem statement by analyzing requirements and the state of the art using techniques like literature research and expert interviews • Define a workplan including tasks, milestones, deliverables and resources and continually assess and modify the plan according to the actual progress of the work (MF1, MF2, MF3, MF4)

	Personal competence: Being a team effort, the project explicitly targets personal competence in terms of: • Working in and managing a team of experts possibly from different academic and cultural backgrounds • Taking part in discussions and learning to contribute the own opinion without overruling other opinions • Self-management and responsibility within the requirements of collaborative work (MKO1, MKO2, MKO3)
Media	Depends on project
Literature	Depends on topic
Methods	Team-discussions, Presentations, Teamwork, Individual preparation of empirical contributions, self-study
Form of assessment	• One or more reports totaling 50-120 pages (depending on the topic) • Presentation (10-90 minutes)
Admission requirements for assessment	• 12-month project: withdrawal within the first 6 weeks possible without failing • 6-month project: withdrawal within the first 3 weeks possible without failing
Duration of assessment	• Final report (length: one or more reports totaling 50-120 pages depending on the topic) • Presentation (10-90 minutes depending on the topic)
Language	English
Offering	HWS/FSS
Person in charge	Professors of the Institute of Computer Science and Business Informatics or of the Area Information Systems of the Business School
Duration of module	1 semester or 2 semesters
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	1st/ 2nd/ 3rd semester

SQ 500: Scientific Research

Module number	SQ 500
Title	Scientific Research
Form of module	Inverted Classroom with Exercises
Type of module	Key Qualification
Level	Master
ECTS	2
Workload	80% (48 hours) self-study and 20% (12 hours) coursework
Prerequisites	-
Aim of module	This course focuses on the basic key competences that are needed to successfully write a scientific paper or a thesis. It is recommended that you take this module before you participate in a seminar. Topics include: • Scientific process and scientific writing • Bibliographic research methodology • Search strategies in bibliographic databases • Finding data for your research • How to read, understand and cite scientific literature • Reference management systems and L^AT_EX
Learning outcomes and qualification goals	Professional competence: • The students understand how to work scientifically and how to write a thesis Methodological competence: • The students can find relevant publications for a research question Personal competence: • Everybody wrote a short overview of his/her research question • Everybody installed and used exemplary tools to support the work process
Media	-
Literature	• <i>The Craft of Research</i> / Wayne C. Booth; Gregory G. Colomb; Joseph M. Williams (Chicago Guides to Writing, Editing, and Publishing); 3rd ed.; University of Chicago Press, 2008; XVII, 317 pages; illustrations; 22cm • L^AT_EX (<i>Wikibook</i>): http://en.wikibooks.org/wiki/LaTeX
Methods	The students can individually work through the video recordings of the lecture and do the accompanying hands-on exercises. The live sessions will be held online as well and are mainly dedicated to clarifying any questions from the students
Form of assessment	Eine digital unterstützte Hausarbeit
Admission requirements for assessment	Successful participation in at least 60% of all online exercises

Duration of assessment	150 minutes
Language	English
Offering	HWS/FSS
Person in charge	Lecturer from the University Library (UB)
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	All semesters possible

Seminar:

CS 701: Selected Topics in Algorithmics and Cryptography

Module number	CS 701
Title	Selected Topics in Algorithmics and Cryptography
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Algorithmics (CS 550) or Cryptography II (CS 651) or Courses in Algorithms or Cryptography or Theoretical Computer Science or Complexity Security at BSc or MSc Level, key qualification scientific research
Aim of module	The students prepare a scientific report on a current research topic on the basis of published papers under guidance of a scientific staff member and give a presentation. The topic will be proposed by the professor, but the students may also propose topics. Active participation in the seminar presentations of fellow students will be expected
Learning outcomes and qualification goals	Professional competence: • The students gain a deep understanding of the research topic, are able to explain the topic in detail in a clean and transparent way and are able to classify the significance of the topic and the results in relation to the current state of research in the corresponding research area Methodological competence: • The students are able to read, understand and explore scientific literature relevant to the topic • They are aware of the need to avoid plagiarism Personal competence: • The student has learned how to find relevant literature for a research topic, write a well-structured and clear report about it and give a presentation • The seminar serves also as preparation for writing and presenting the master's thesis
Media	Scientific papers and books, Presentation systems like PowerPoint or projector L ^A T _E X
Literature	Depends on the topic
Methods	Do scientific work independently under the guidance of a research staff member and manage an active discussion on the topic in a group of peers

Form of assessment	• Presentation and discussion (15-60 minutes) • Written essay/report (5-25 pages) • Participation
Admission requirements for assessment	Timely hand-in of seminar papers and presentation materials
Duration of assessment	• 60 minutes presentation with 15-30 minutes discussion • Written report (length: 15 pages)
Language	English
Offering	FSS
Person in charge	Prof. Dr. Matthias Krause
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Mathematics in Business and Economics
Semester	3rd semester

CS 704: Master Seminar Artificial Intelligence

Module number	CS 704
Title	Master Seminar Artificial Intelligence
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Decision Support or Data Mining or Knowledge Management
Aim of module	The student prepares a scientific paper and gives a presentation on a current research topic based on published research. State-of-the-art topics are proposed by the professors. The paper and the presentation are prepared under the guidance of a professor or a research staff member. Active participation in the seminar discussions is expected
Learning outcomes and qualification goals	Professional competence: • The student gains a deep understanding of the research topic • He/she is able to describe/summarize the topic in detail in his/her own words • He/she reflects on the topic and judges the contribution of the research papers (MK1, MK2, MK3) Methodological competence: • The student is able to write a well-structured scientific paper and to present his/her results • He/she is also aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar (MF2) Personal competence: • The student has learned how to write a well-structured, concise paper and give a presentation • This is part of the preparation to write and present a master's thesis
Media	Scientific papers and books, final presentation with L ^A T _E X slides
Literature	Depends on the topic of the seminar
Methods	Do scientific reading independently under the guidance of a professor or a research staff member, active discussions in a group of peers
Form of assessment	We offer two types of seminars: • <i>Seminar with a focus on theory and literature:</i> Two presentations (30 minutes overall) and a final seminar report (15-20 pages) • <i>Practical Seminar:</i> Two presentations (30 minutes overall) and a final seminar report (12-15 pages) presenting own results

	Offered in exceptional cases as group work. In such cases, the practical task is more extensive/challenging. Code has to be submitted in addition to the final report. More about content and type of the seminar can be found at https://www.uni-mannheim.de/dws/teaching/course-details/courses-for-master-candidates/
Admission requirements for assessment	-
Duration of assessment	• Both presentations must be given within a span of 5-10 weeks • Final report has to be submitted after this period
Language	English
Offering	Irregular
Person in charge	Prof. Dr. Heiner Stuckenschmidt
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	3rd semester

CS 707: Data Analytics Seminar

Module number	CS 707
Title	Data Analytics Seminar
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	• CS Fundamentals • Relevant Master Courses (depending on the topic) • Key qualification Scientific Research
Aim of module	Explore, present and critically discuss recent research literature in the areas of machine learning and/or data processing. The precise topics vary from term to term
Learning outcomes and qualification goals	Professional competence: • The student gains a deep understanding of the research topic Methodological competence: • The student is able to read, understand, explore, present, defend and critically discuss scientific literature relevant to his/her topic • He/she is also aware of the need to avoid plagiarism Personal competence: • The seminar fosters critical thinking and discussion skills • Students will be prepared to research and critically assess current scientific literature in preparation of their Master's thesis
Media	Scientific papers and books, software and datasets, presentations
Literature	Depends on the topic of the seminar
Methods	Do scientific work independently under the guidance of a professor or a research staff member, active discussions in a group of peers
Form of assessment	Individual grading of the oral presentations, experimental results (if applicable), active participation in the seminar's discussion rounds, a written discussion summary, and timeliness of hand-ins
Admission requirements for assessment	-
Duration of assessment	N/A
Language	English
Offering	HWS/FSS (typically)
Person in charge	Prof. Dr. Rainer Gemulla
Duration of module	1 semester

Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	3rd semester

CS 708: Seminar Software Engineering

Module number	CS 708
Title	Seminar Software Engineering
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Bachelor's degree, the fundamentals
Aim of module	Student prepares a scientific paper and gives a presentation on a current software engineering research topic based on published research papers. State-of-the-art topics are proposed by the software engineering group. Active participation in the seminar discussions is expected
Learning outcomes and qualification goals	Professional competence: • The student gains a deep understanding of the research topic • They are able to describe/summarize the topic in detail in their own words • They reflect on the topic and judge the contribution of the research papers Methodological competence: • The student is able to find the relevant literature for their topic, to write a well-structured scientific paper and to present their results • They are also aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar Personal competence: • The student has learned how to find relevant literature for a research topic, write a well-structured, concise paper about it and give a presentation
Media	Scientific papers and books, presentation
Literature	Depends on the topic of the seminar
Methods	Scientific work performed independently under the guidance of a member of the software engineering group
Form of assessment	• Seminar paper (final report): 12-13 pages excluding references • Presentation (15 minutes) plus discussion (5-10 minutes) • Peer review: 1-5 pages • Accompanying work products (e.g., code, models, spreadsheets, documents etc.)
Admission requirements for assessment	-

Duration of assessment	Presentation (15 minutes) plus discussion (5-10 minutes)
Language	English
Offering	Irregular
Person in charge	Prof. Dr. Colin Atkinson
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	1st/ 2nd/ 3rd semester

CS 709: Seminar Text Analytics

Module number	CS 709
Title	Seminar Text Analytics
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	IS 661 (IS 6610) “Text Analytics” or IE 663 “Web Search and Information Retrieval” or IE 696 “Advanced Text Analytics”, Fundamental notions of linear algebra and probability theory
Aim of module	In this seminar, students write a survey/scientific paper and provide an overview presentation of state-of-the-art research, as found within the existing literature (i.e., published research papers). Topics of interest focus around a variety of problems and tasks from the fields of Natural Language Processing and Information Retrieval. The paper and the presentation are prepared under the guidance of a professor or a research staff member
Learning outcomes and qualification goals	Professional competence: • Students will acquire a deep understanding of the research topic • They are expected to describe in-depth and summarize the topic in detail in their own words, as well as to judge the contribution of the research papers to ongoing research Methodological competence: • Students will develop methods and skills to find relevant literature for their topic, to write a well-structured survey/scientific paper and to present their results • They will also be aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar Personal competence: • Students will acquire skills on how to find relevant literature for a research topic, write a well-structured, concise paper about it and present the results of their work • They are well prepared to write and present a master’s thesis
Media	Scientific papers and books, presentation with PowerPoint or L ^A T _E X
Literature	Depends on the topic of the seminar
Methods	Do scientific work independently under the guidance of a professor or a research staff member, active discussions in a group of peers

Form of assessment	• Flash presentation (3 minutes) • Final presentation (15 minutes presentation + discussion + moderation) • Report (10 pages + references/appendix) • Peer review • Optionally: code hand-in
Admission requirements for assessment	-
Duration of assessment	-
Language	English or German
Offering	Irregular
Person in charge	Prof. Dr. Simone Paolo Ponzetto
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	3rd semester

CS 710: Selected Topics in Data Science

Module number	CS 710
Title	Selected Topics in Data Science
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Bachelor's degree, the fundamentals
Aim of module	In this seminar, students perform scientific research, either in the form of a literature review or by conducting a small experiment, or a mixture of both, and prepare a written report about the results. Topics of interest focus around a variety of problems and tasks from the fields of Data Mining, Web Mining, or the Semantic Web
Learning outcomes and qualification goals	Professional competence: • Students will acquire a deep understanding of the research topic • He/she is expected to describe in-depth and summarize the topic in detail in his/her own words, as well as to judge the contribution of the research papers to ongoing research (MK1, MK2, MK3) Methodological competence: • Students will develop methods and skills to find relevant literature for his/her topic, to prepare methodologically sound scientific experiments, and to write a well-structured scientific paper and to present his/her results • He/she will also be aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar (MF2) Personal competence: • Students will acquire skills on how to find relevant literature for a research topic, organize a small research task, write a wellstructured, concise paper about it and present the results of their work • He/she is well prepared to write and present a master's thesis (MF2)
Media	Scientific papers and books
Literature	Depends on the topic of the seminar
Methods	Do scientific work independently under the guidance of a professor or a research staff member
Form of assessment	• Seminar paper (5-25 pages depending on the topic) • Peer Review (1-10 pages depending on the topic) • Presentation (15-60 minutes depending on the topic)
Admission requirements for assessment	-

Duration of assessment	• 6-8 weeks (writing of the essay) • 15-60 minutes (presentation)
Language	English or German
Offering	Irregular
Person in charge	Prof. Dr. Heiko Paulheim
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	3rd semester

CS 715: Large-Scale Data Integration Seminar

Module number	CS 715
Title	Solving Complex Tasks using Large Language Models
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Bachelor's degree, the fundamentals
Aim of module	The seminar explores prompt engineering, fine-tuning, as well as agent-based techniques for enabling LLMs to handle complex tasks as well as using LLMs to evaluate complex outputs. The seminar features literature as well as experimental topics. The goal of the literature topics is to summarize the state of the art concerning the application and evaluation of LLMs for a specific task. The goal of the experimental topics is to verify the utility of advanced techniques by applying them to tasks beyond the tasks used in the original papers
Learning outcomes and qualification goals	Professional competence: • Students will acquire a deep understanding of the state of the art with respect to their research topic • They learn to summarize the state of the art, judge methods along different criteria, design experiments, and present experimental results Methodological competence: • Students will develop methods and skills to find relevant literature for their topic, to prepare methodologically sound scientific experiments and to write a well-structured scientific paper and to present their results • They will also be aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar Personal competence: • Students will acquire skills on how to find relevant literature for a research topic, organize a small research task, write a wellstructured, concise paper about it and present the results of their work
Media	Scientific papers and books
Literature	Depends on the specific topic
Methods	Perform scientific work independently under the guidance of a professor or a research staff member
Form of assessment	• Report (12-15 pages excluding references) • Presentation (12 minutes + 8 minutes discussion)

Admission requirements for assessment	-
Duration of assessment	6-8 weeks (writing of the seminar paper)
Language	English
Offering	Irregular
Person in charge	Prof. Dr. Christian Bizer
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, M.Sc. Social Data Science, Teacher Education Informatics
Semester	3rd semester

CS 716: IT-Security

Module number	CS 716
Title	IT-Security
Form of module	Block seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Bachelor's degree, the fundamentals
Aim of module	Seminar participants choose a topic from a given list of topics or, if applicable, based on their own suggestions. They prepare a presentation on this topic and write a corresponding paper. They also conduct literature research and systemize the current state of research. The focus here is on working out and understanding the scientific content and being able to classify and evaluate it. All participants give a presentation. Active participation in the seminar is expected
Learning outcomes and qualification goals	Professional competence: • Students have in-depth knowledge in the field covered by the seminar Methodological competence: • Students are able to find relevant, current literature on a research topic, thoroughly understand a research topic, process it both in terms of content and didactics, and present it to others. They will be well prepared to write and present a Master's Thesis Personal competence: • Students are able to work independently • They have learned how to find relevant literature on a research topic, delve deeper into a subject, write a well-structured, concise report on it, and give a presentation
Media	Scientific papers and books, final presentation with PowerPoint or similar software
Literature	Depends on the topic of the seminar
Methods	Independent academic work and preparation of a presentation (lecture and paper) in collaboration with a professor or research assistant. Active discussions in a group of fellow students
Form of assessment	• Presentation • Technical discussion • Seminar paper (8 pages, double-column)
Admission requirements for assessment	Timely hand-in of seminar paper and presentation
Duration of assessment	• 90 minutes for presentation and technical discussion • Seminar paper (8 pages, double-column)

Language	English
Offering	Irregular
Person in charge	Prof. Dr. Frederik Armknecht
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, Teacher Education Informatics, M.Sc. Mathematics in Business and Economics
Semester	3rd semester

CS 717: Master Seminar on Computer Vision

Module number	CS 717
Title	Master Seminar on Computer Vision
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Higher Level Computer Vision, Generative Computer Vision Models or Image Processing
Aim of module	The student prepares a scientific paper and gives a presentation on a current research topic based on published research. State-of-the-art topics are proposed by the professors. The paper and the presentation are prepared under the guidance of a professor or a research staff member. Active participation in the seminar discussions is expected
Learning outcomes and qualification goals	Professional competence: • The student gains a deep understanding of the research topic • He/she is able to describe/summarize the topic in detail in his/her own words • He/she reflects on the topic and judges the contribution of the research papers Methodological competence: • The student is able to write a well-structured scientific paper and to present his/her results • He/she is also aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar Personal competence: • The student has learned how to write a well-structured, concise paper and give a presentation • This is part of the preparation to write and present a master's thesis
Media	Scientific papers and books
Literature	Depending on the topic of the seminar
Methods	Do scientific reading independently under the guidance of a professor or a research staff member, active discussions in a group of peers
Form of assessment	• 50% Presentations (one 3-5 minutes presentation and one 25-30 minutes presentation) • 50% Seminar Report (10-20 pages, has to be submitted after three month)
Admission requirements for assessment	-

Duration of assessment	N/A
Language	English
Offering	HWS/FSS
Person in charge	Prof. Dr.-Ing. Margret Keuper
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	3rd semester

CS 718: AI and Data Science in Fiction and Society

Module number	CS 718
Title	AI and Data Science in Fiction and Society
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Bachelor's degree
Aim of module	In this seminar, students analyze and discuss fictional works in the area of AI and data science with respect to technological and societal aspects. The present the results orally and in a written report
Learning outcomes and qualification goals	Professional competence: - Students will learn about societal effects of AI and data science and become aware of potential threats and dangers, but also of chances of those new technologies (MK1) Methodological competence: - Students will develop methods and skills to find relevant literature for his/her topic, and to write a well-structured scientific paper and to present his/her results - He/she will be also aware of the need to avoid plagiarism - The key qualification Scientific Research is highly recommended as a prerequisite for the seminar (MF1, MF2, MF3) Personal competence: - Students will acquire skills on how to find relevant literature for a research topic, discuss a fictional work using secondary literature as background material, write a well-structured, concise paper about it and present the results of their work - He/she is well prepared to write and present a master's thesis (MKO2)
Media	Fictional and non-fictional texts
Literature	A detailed literature list is compiled for each offering
Methods	Do scientific work independently under the guidance of a professor or a research staff member
Form of assessment	- Seminar paper (15-20 pages depending on the topic) - Peer Review (up to 10 pages) - Presentation (20-30 minutes depending on the topic)
Admission requirements for assessment	-
Duration of assessment	6-8 weeks (writing of the essay) 20-30 minutes (presentation)

Language	English
Offering	HWS
Person in charge	Prof. Dr. Heiko Paulheim
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Data Science, M.Sc. Business Informatics, Teacher Education Informatics
Semester	3rd semester

CS 720: Uncertainty Estimation

Module number	CS 720
Title	Uncertainty Estimation
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Bachelor's degree, the fundamentals
Aim of module	In this seminar, students perform scientific research, either in the form of a literature review or by conducting a small experiment, or a mixture of both, and prepare a written report about the results. Topics of interest focus around a variety of problems and tasks from the fields of Data Mining, Web Mining, or the Semantic Web
Learning outcomes and qualification goals	Professional competence: • Students will acquire a deep understanding of the research topic • He/she is expected to describe in-depth and summarize the topic in detail in his/her own words, as well as to judge the contribution of the research papers to ongoing research (MK1, MK2, MK3) Methodological competence: • Students will develop methods and skills to find relevant literature for his/her topic, to prepare methodologically sound scientific experiments, and to write a well-structured scientific paper and to present his/her results • He/she will also be aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar (MF2) Personal competence: • Students will acquire skills on how to find relevant literature for a research topic, organize a small research task, write a wellstructured, concise paper about it and present the results of their work • He/she is well prepared to write and present a master's thesis (MF2)
Media	Scientific papers and books
Literature	Depends on the topic of the seminar
Methods	Do scientific work independently under the guidance of a professor or a research staff member
Form of assessment	• Seminar paper (5-25 pages depending on the topic) • Peer Review (1-10 pages depending on the topic) • Presentation (15-60 minutes depending on the topic)
Admission requirements for assessment	-

Duration of assessment	• 6-8 weeks (writing of the report) • 15-60 minutes (presentation)
Language	English or German
Offering	Irregular
Person in charge	Prof. Dr. Heiko Paulheim
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science, Teacher Education Informatics
Semester	3rd semester

CS 721: Seminar Data-Science I

Module number	CS 721
Title	Seminar Data-Science I
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	There are no formal requirements. However, previous participation in the courses “Network Science” and “Text Analytics” are recommended
Aim of module	In this seminar, students perform scientific research, either in the form of a literature review or by conducting a small experiment, or a mixture of both, and prepare a written report about the results. Topics of interest focus around a variety of problems and tasks from the fields of Data-Science, Network Science and Text Mining
Learning outcomes and qualification goals	Professional competence: • Students will acquire a deep understanding of the research topic • He/she is expected to describe in-depth and summarize the topic in detail in his/her own words, as well as to judge the contribution of the research papers to ongoing research (MK1, MK2, MK3) Methodological competence: • Students will develop methods and skills to find relevant literature for his/her topic, to prepare methodologically sound scientific experiments, and to write a well-structured scientific paper and to present his/her results • He/she will also be aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar (MF2) Personal competence: • Students will acquire skills on how to find relevant literature for a research topic, organize a small research task, write a wellstructured, concise paper about it and present the results of their work • He/she is well prepared to write and present a master’s thesis (MF2)
Media	Slides, scientific papers, blackboard (electronic)
Literature	Depends on the topic
Methods	Self-study of assigned material, presentation of scientific articles, joint discussion of work, collaboration with peers
Form of assessment	• Preparation of a seminar report (12-14 pages, plus bibliography) • Presentation of the seminar paper and answering questions (40 minutes) • Participation in the presentation discussions • Reflection on a seminar report written by someone else

Admission requirements for assessment	-
Duration of assessment	-
Language	English
Offering	Irregular
Person in charge	Prof. Markus Strohmaier
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	3rd semester

CS 730: Advanced Implementation Techniques for Database Systems

Module number	CS 730
Title	Advanced Implementation Techniques for Database Systems
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	DBSI, DBSII, Query Optimization
Aim of module	The student prepares a scientific paper and gives a presentation on a current research topic based on published research. State-of-the-art topics are proposed by the professors. The paper and the presentation are prepared under the guidance of a professor or a research staff member. Active participation in the seminar discussions is expected
Learning outcomes and qualification goals	Professional competence: • The student gains a deep understanding of the research topic • They are able to describe/summarize the topic in detail in their own words. They reflect on the topic and judges the contribution of the research paper Methodological competence: • The student is able to write a well-structured scientific paper and to present their results • They are also aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar Personal qualification: • The student has learned how to write a well-structured, concise paper and give a presentation • This is part of the preparation to write and present a master's thesis
Media	Slides, blackboard
Literature	To be announced in the seminar
Methods	Do scientific work independently under the guidance of a professor or a research staff member. Active discussions in a group of peers
Form of assessment	• Presentation (80%) • Report (20%)
Admission requirements for assessment	-
Duration of assessment	• Presentation (30 minutes) • Report (17 pages)

Language	English
Offering	Irregular
Person in charge	Prof. Dr. Guido Moerkotte
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics
Semester	2nd/ 3rd semester

CS 731: Database Theory

Module number	CS 731
Title	Database Theory
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Theoretical Computer Science, Complexity Theory, Decidability Theory, DBSI
Aim of module	The student prepares a scientific paper and gives a presentation on a current research topic based on published research. State-of-the-art topics are proposed by the professors. The paper and the presentation are prepared under the guidance of a professor or a research staff member. Active participation in the seminar discussions is expected
Learning outcomes and qualification goals	Professional competence: • The student gains a deep understanding of the research topic • They are able to describe/summarize the topic in detail in their own words • They reflect on the topic and judges the contribution of the research paper Methodological competence: • The student is able to write a well-structured scientific paper and to present their results • They are also aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar Personal qualification: • The student has learned how to write a well-structured, concise paper and give a presentation • This is part of the preparation to write and present a master's thesis
Media	Slides, blackboard
Literature	To be announced in the seminar
Methods	Do scientific work independently under the guidance of a professor or a research staff member. Active discussions in a group of peers
Form of assessment	• Presentation (80%) • Report (20%)
Admission requirements for assessment	-

Duration of assessment	• Presentation (30 minutes) • Report (17 pages)
Language	English
Offering	Irregular
Person in charge	Prof. Dr. Guido Moerkotte
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics
Semester	2nd/ 3rd semester

CS 733: Advanced Topics in Process Mining

Module number	CS 733
Title	Advanced Topics in Process Mining
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Participants are expected to have fundamental knowledge of process mining, including basic concepts, techniques, and tools, to actively engage with the seminar content
Aim of module	In this seminar, each student conducts scientific research on advanced topics in process mining, either as a literature review, a small experimental study, or a combination of both. Each student prepares a written report and delivers a presentation to the class, sharing their findings and insights with peers. Topics of interest include classical process mining tasks, such as process discovery, conformance checking, and predictive process monitoring, explored through current research trends and novel approaches, including generative AI, machine learning applications, and the analysis of non-traditional event data types. The seminar encourages students to investigate innovative methods, emerging challenges, and practical implications in the field
Learning outcomes and qualification goals	Professional competence: • Students will develop a deep understanding of the research topic • They are expected to critically analyze and comprehensively summarize the topic in their own words, as well as evaluate the contribution of the research papers to ongoing research in the field Methodological competence: • Students will develop the methods and skills needed to identify relevant literature for their topic, design methodologically sound scientific experiments, write a well-structured scientific paper, and effectively present their results • They will also gain awareness of the importance of academic integrity and the need to avoid plagiarism Personal qualification: • Students will acquire the skills required to identify relevant literature for a research topic, organize and conduct a small-scale research project, write a well-structured and concise academic paper, and effectively present their results • They will be well prepared to write and present a Master's thesis
Media	Scientific papers and books

Literature	• Depends on the specific topic • For a general overview of the field, the following book is recommended: Wil van der Aalst: <i>Process Mining: Data Science in Action</i>, 2nd edition
Methods	Perform scientific work independently under the guidance of a professor or a research staff member
Form of assessment	• Seminar paper (50%) • Mid-term presentation (10%) • Final presentation (40%)
Admission requirements for assessment	-
Duration of assessment	• Seminar paper: 18 pages (excluding appendix and references) • Mid-term presentation: 10 minutes presentation + 10 minutes discussion • Final presentation: 20 minutes presentation + 10 minutes discussion
Language	English
Offering	Irregular
Person in charge	Prof. Dr. Daniel Schuster
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics, M.Sc. Data Science
Semester	3rd semester

IS 712 (IS 7120): Contemporary Issues in Information Systems Research

Module number	IS 712 (IS 7120)
Title	Contemporary Issues in Information Systems Research
Form of module	Seminar
Type of module	Seminar
Level	Master
ECTS	4
Workload	120 h per semester
Prerequisites	Bachelor's degree, the fundamentals
Aim of module	The primary objective of the seminar is to analyze information systems development and use from different perspectives. The secondary objective is to provide students with techniques of scientific writing in a fascinating real-world setting. Important aspects are the evaluation, structuring, and classification of existing research work and the presentation of a detailed and thorough overview of the current state of the art. In addition, scientific work also includes the creation of new knowledge. The participation in the seminar can be regarded as an important preliminary step towards a successful completion of the final thesis
Learning outcomes and qualification goals	Professional competence: • The student gains a deep understanding of the research topic • They are able to conduct basic scientific research • They reflect on the topic and judges the contribution of the research papers Methodological competence: • The student is able to find the relevant literature for their topic, to write a well-structured scientific paper and to present their results • They are also aware of the need to avoid plagiarism • The key qualification Scientific Research is highly recommended as a prerequisite for the seminar Personal qualification: • The student has learned how to find relevant literature for a research topic, write a well-structured, concise paper about it and give a presentation • They are well prepared to write and present a master's thesis
Media	Scientific papers and books; final presentation with PowerPoint
Literature	Depends on the topic of the seminar
Methods	Do scientific work independently under the guidance of a professor or a research staff member. Active discussions in a group of peers

Form of assessment	• Preparation of a seminar paper (15-25 pages written, plus bibliography) • Presentation of the seminar paper and answering questions (15-30 minutes) • Reflection on a seminar paper written by someone else (discussant statement; 2-5 minutes) • Participation in the plenary discussion
Admission requirements for assessment	-
Duration of assessment	N/A
Language	English
Offering	Irregular
Person in charge	Dr. Monica Fallon
Duration of module	1 semester
Further modules	-
Programs	Mannheim Master in Management, M.Sc. Business Informatics
Semester	3rd semester

E. Master Thesis

MA 650: Master Thesis

Module number	MA 650
Title	Master's Thesis
Form of module	Master's Thesis
Type of module	Thesis
Level	Master
ECTS	30
Workload	Self-study: 840 h per semester
Prerequisites	The student is required to have obtained at least 60 ECTS credits in order to register for his or her master thesis
Aim of module	Develop a deep understand of an advanced topic of business informatics or computer science
Learning outcomes and qualification goals	Professional competence: • The student has a deep understanding of an advanced topic (MK1) Methodological competence: • The student is familiar with methods for analysing and independently solving advanced, complex problems (MK1, MK2, MK3) Personal competence: • The student has the capability to understand, analyse and independently find solutions to advanced, complex problems • The student has the capability to assess and understand the state-of-the-art in business informatics and adapt the latest technologies and methods to solve real world problems • The student is able to present a complex topic in written and oral form in a clear and understandable way (MF1, MF2, MF3, MF4, MKO2, MKO3)
Media	Various
Literature	Topic dependent
Methods	Independent research work
Form of assessment	Written thesis, 10-120 pages in length depending on the topic
Admission requirements for assessment	To be permitted to write the master thesis, the student is to obtain at least 60 ECTS
Duration of assessment	6 months (execution of the thesis work and writing of the thesis manuscript)
Language	English or German
Offering	HWS/FSS

Person in charge	Professors of the Institute of Computer Science and Business Informatics or of the Area Information Systems of the Business School
Duration of module	1 semester
Further modules	-
Programs	M.Sc. Business Informatics
Semester	4th semester

Abbreviations

Knowledge

The courses are divided into two groups - fundamental courses and specialization courses. After studying mandatory fundamental courses in computer science and business administration, in their advanced courses students can focus on the concepts and methods of computer science, the application of these methods in system design and development, or on the use of information technology in business processes. In addition to the regular lecture courses, students participate in a team project.

During their studies

- (MK1) all students develop a deep understanding of the relevant concepts, methods and problem-solving strategies used in different application domains;
- (MK2) technology-oriented students learn the concepts, algorithms and strategies used to solve concrete, practical application-oriented problems in business informatics;
- (MK3) business-oriented students develop a deep understanding of how to deploy, develop and manage information systems.

As part of this education, students become familiar with a wide range of models, modelling languages, methods, and tools. Students also learn how to collect, structure, manipulate, prepare, communicate, and use data, information, and knowledge to define and control processes in companies and industrial scenarios.

Capabilities

After completing their studies, students have the ability to

- (MF1) apply a wide range of abstraction and analysis techniques;
- (MF2) understand, interpret, describe, and present relevant scientific publications;
- (MF3) exploit the latest scientific results;
- (MF4) independently tackle problems in business informatics and describe their results in a structured, written form;
- (MF5) continue their studies at the PhD level, if their results are of sufficient quality.

Competencies

After completing their studies, students have the competences needed to

- (MKO1) apply their knowledge and capabilities to solve specific problems in a team context;
- (MKO2) use their interdisciplinary education to mediate between technical and non-technical individuals;
- (MKO3) to evaluate the latest changes in programming languages, systems, business models and process models and, wherever possible, exploit them to develop better solutions to business informatics problems.